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Analyses to support the Dolphinfish Management Strategy Evaluation in the U.S. Atlantic

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National Oceanic and Atmospheric Administration
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Northwest Fisheries Science Center

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1 Introduction

Here we will be displaying various data summaries and analyses related to dolphin management in the U.S. South Atlantic.

1.1 Details

The documentation of this data exploration was developed in R, using R Studio and Quarto.

2 NOAA Fisheries recreational landings data

Here we will partition recreational landings data from NOAA Fisheries' Marine Recreational Information Program (MRIP), a standardized, peer-reviewed suite of recreational fishing surveys used to produce catch and effort estimates for quota monitoring and stock assessment in the United States. MRIP is a standardized survey suite that employs a complex statistical sampling design stratified across time, regions of the coast, and different fishing modes, which allows the intercept data to be scaled up by effort estimates to calculate total landings.

2.1 Data access and upload

Data are downloaded directly from the Miami server using the files provided to SERO for ACL monitoring. The current location is: M://SFD/SECM-SFD/ACL/2025_Aug19_-MRIP_FES/. The landings are in FES MRIP and the units are in pounds whole weight.

```
# clear workspace
rm(list = ls())

# read in data file
load("data/mrip_fes_rec81_25wv2_23June25.RData")  # most recent file on SFD server
head(dat)
```

```
  REC_ACL_MRIP_FES_ID YEAR MONTH WAVE SUB_REG NEW_ST NEW_STA NEW_MODE NEW_MODEN
1            73407425 1989    NA    4       6      9      NC        4      Priv
2            73407426 1989    NA    4       6      9      NC        4      Priv
3            73407427 1989    NA    4       6      9      NC        4      Priv
4            73407428 1989    NA    4       6      9      NC        4      Priv
5            73407429 1989    NA    4       6      9      NC        4      Priv
6            73407430 1989    NA    4       6      9      NC        4      Priv
  AREA_X NEW_AREA  NEW_AREAN FL_REG NC_REG   DS              NEW_SCI
1      5        5    Inshore     NA      S MRIP Centropristis striata
2      5        5    Inshore     NA      N MRIP Centropristis striata
```

```
3      2        2  Ocean>3mi     NA      N MRIP Centropristis striata
4      2        2  Ocean>3mi     NA      S MRIP Centropristis striata
5      1        1 Ocean<=3mi     NA      N MRIP Centropristis striata
6      1        1 Ocean<=3mi     NA      S MRIP Centropristis striata
        NEW_COM    SP_CODE SPECIES_ITIS SPECIES        AB1          B2
1 black sea bass 8835020301       167687         34181.4033 121019.1368
2 black sea bass 8835020301       167687           795.8612   4445.0527
3 black sea bass 8835020301       167687           549.5390   2919.4840
4 black sea bass 8835020301       167687         71823.2390  24548.2535
5 black sea bass 8835020301       167687           415.7642     87.5928
6 black sea bass 8835020301       167687         48728.5632  11859.6428
           A         B1 LBSEST_SECWWT LBSEST_SECSOURCE SAMPLE_SIZE_USED
1 31761.4755  2419.9278    11457.8425          srysmwa               41
2   584.4155   211.4457      266.7782          srysmwa               41
3   549.5390     0.0000      484.4319          srysmwa              119
4 33343.1844 38480.0546    63313.9276          srysmwa              119
5   415.7642     0.0000      294.0623          srysmwa               71
6 40585.0152  8143.5480    34464.8156          srysmwa               71
   AVG_LEN   AVG_WGT SRYSMWA_AVG_LEN SRYSMWA_AVG_WGT SRYSMWA_SAMPLE_SIZE
1 203.4572 0.3352069        203.4572       0.3352069                  41
2 203.4572 0.3352069        203.4572       0.3352069                  41
3 296.2590 0.8815243        296.2590       0.8815243                 119
4 296.2590 0.8815243        296.2590       0.8815243                 119
5 261.9949 0.7072816        261.9949       0.7072816                  71
6 261.9949 0.7072816        261.9949       0.7072816                  71
  SRYSMWA_FLAGGED SRYSMW_AVG_LEN SRYSMW_AVG_WGT SRYSMW_SAMPLE_SIZE
1              NA       269.2563      0.7310037                231
2              NA       269.2563      0.7310037                231
3              NA       269.2563      0.7310037                231
4              NA       269.2563      0.7310037                231
5              NA       269.2563      0.7310037                231
6              NA       269.2563      0.7310037                231
  SRYSMW_FLAGGED SRYSM_AVG_LEN SRYSM_AVG_WGT SRYSM_SAMPLE_SIZE SRYSM_FLAGGED
1             NA      259.7694     0.6948228               556            NA
2             NA      259.7694     0.6948228               556            NA
3             NA      259.7694     0.6948228               556            NA
4             NA      259.7694     0.6948228               556            NA
5             NA      259.7694     0.6948228               556            NA
6             NA      259.7694     0.6948228               556            NA
  SRYS_AVG_LEN SRYS_AVG_WGT SRYS_SAMPLE_SIZE SRYS_FLAGGED SRY_AVG_LEN
1      270.955    0.8099641              982           NA    269.1276
```

```
2      270.955    0.8099641              982           NA    269.1276
3      270.955    0.8099641              982           NA    269.1276
4      270.955    0.8099641              982           NA    269.1276
5      270.955    0.8099641              982           NA    269.1276
6      270.955    0.8099641              982           NA    269.1276
  SRY_AVG_WGT SRY_SAMPLE_SIZE SRY_FLAGGED SR_AVG_LEN SR_AVG_WGT SR_SAMPLE_SIZE
1   0.8085211            1703          NA   303.6871   1.050403          46055
2   0.8085211            1703          NA   303.6871   1.050403          46055
3   0.8085211            1703          NA   303.6871   1.050403          46055
4   0.8085211            1703          NA   303.6871   1.050403          46055
5   0.8085211            1703          NA   303.6871   1.050403          46055
6   0.8085211            1703          NA   303.6871   1.050403          46055
  SR_FLAGGED S_AVG_LEN S_AVG_WGT S_SAMPLE_SIZE S_FLAGGED     VAR_B2   VAR_AB1
1         NA  344.4373  1.364082        305089        NA 4072492794 310099094
2         NA  344.4373  1.364082        305089        NA    2866289    258947
3         NA  344.4373  1.364082        305089        NA          0         0
4         NA  344.4373  1.364082        305089        NA  105731413 534702898
5         NA  344.4373  1.364082        305089        NA          0         0
6         NA  344.4373  1.364082        305089        NA   22265768 286196169
         SA_LABEL GOM_LABEL REC_ACL MIGRA_GRP AGG_MODEN JURISDICTION PS_REG
1 Snapper Grouper                 1             Private        State      6
2 Snapper Grouper                 1             Private        State      5
3 Snapper Grouper                 1             Private      Federal      5
4 Snapper Grouper                 1             Private      Federal      6
5 Snapper Grouper                 1             Private        State      5
6 Snapper Grouper                 1             Private        State      6
  COUNCILREG GFMC_FMP SAFMC_FMU LBSEST_SECGWT AGGR_AREA AREA FIRST_MONTH
1          6                  Y     9710.0360             NA          NA
2          5                  Y      226.0832             NA          NA
3          5                  Y      410.5355             NA          NA
4          6                  Y    53655.8708             NA          NA
5          5                  Y      249.2054             NA          NA
6          6                  Y    29207.4709             NA          NA
  LAST_MONTH ALT_FLAG CHTS_CL CHTS_H CHTS_RL CHTS_WAB1C CHTS_WAB1H MODE_FX
1         NA        0       0      0       0          0          0       7
2         NA        0       0      0       0          0          0       7
3         NA        0       0      0       0          0          0       7
4         NA        0       0      0       0          0          0       7
5         NA        0       0      0       0          0          0       7
6         NA        0       0      0       0          0          0       7
  NEW_SEASN SEASON
```

```
1                NA
2                NA
3                NA
4                NA
5                NA
6                NA
```

```
#apply(dat[2:18], 2, table, useNA = "always")

# confirm nomenclature for dolphin and subset by species --------------------------
table(dat$NEW_COM[grep("dolphin", dat$NEW_COM)])
```

```
dolphin
   9377
```

```
table(dat$NEW_SCI[grep("dolphin", dat$NEW_COM)])
```

```
Coryphaena hippurus
               9377
```

```
d <- dat[which(dat$NEW_COM == "dolphin"), ]
#apply(d[2:18], 2, table, useNA = "always")
d <- d[which(d$YEAR < 2025), ]    # take out 2025 because incomplete year
```

2.2 Separate Gulf and Atlantic recreational landings

Now we filter the data set to separate out the Gulf versus Atlantic landings.

```
# codes for WFL regions: 1 (NW FL Panhandle- Escambia to Dixie co.), 2 (SW FL Peninsula- Le
table(d$FL_REG, useNA = "always")
```

```
   1    2    3    4    5 <NA>
 387  110  568 1022  312 6943
```

```r
# subregional codes: 6 is Atlantic and 7 is Gulf
table(d$FL_REG, d$SUB_REG, useNA = "always")
```

```
          4    5    6    7 <NA>
  1       0    0    0  387    0
  2       0    0    0  110    0
  3       0    0    0  568    0
  4       0    0 1022    0    0
  5       0    0  312    0    0
  <NA>   96  585 3836 2426    0
```

```r
unique(d$NEW_STA)
```

```
 [1] "NC"     "FLE"    "FLW"    "SC"     "GA"     "MS"     "AL"     "LA"
 [9] "TX"     "FLE/GA" "RI"     "VA"     "DE"     "MD"     "NJ"     "CT"
[17] "MA"     "NY"     "LA/MS"  "AL/FLW"
```

```r
lis <- c("TX", "LA", "LA/MS", "MS", "AL", "AL/FLW")  # filter out the Gulf states
d1 <- d[-which(d$NEW_STA %in% lis), ]
unique(d1$NEW_STA)
```

```
 [1] "NC"     "FLE"    "FLW"    "SC"     "GA"     "FLE/GA" "RI"     "VA"
 [9] "DE"     "MD"     "NJ"     "CT"     "MA"     "NY"
```

```r
table(d1$FL_REG, d1$NEW_STA, useNA = "always")
```

```
         CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
  1       0    0    0      0  387    0    0    0    0    0    0    0    0    0
  2       0    0    0      0  110    0    0    0    0    0    0    0    0    0
  3       0    0    0      0  568    0    0    0    0    0    0    0    0    0
  4       0    0 1022      0    0    0    0    0    0    0    0    0    0    0
  5       0    0  312      0    0    0    0    0    0    0    0    0    0    0
  <NA>   14  114    0   1794  174   54   20  125 1344  108   73   62  644  165

       <NA>
```

```
  1        0
  2        0
  3        0
  4        0
  5        0
  <NA>     0
```

```r
dim(d1)
```

```
[1] 7090   83
```

```r
d1 <- d1[-which(d1$FL_REG == 1 | d1$FL_REG == 2), ]   # filter out FL regions 1 and 2 but r
dim(d1)
```

```
[1] 6593   83
```

```r
d1 <- d1[-which(is.na(d1$FL_REG) & d1$NEW_STA == "FLW"), ]  # filter out FLW with unknown F
dim(d1)
```

```
[1] 6419   83
```

```r
table(d1$FL_REG, d1$NEW_STA, useNA = "always")  # retained are all Atlantic states plus reg
```

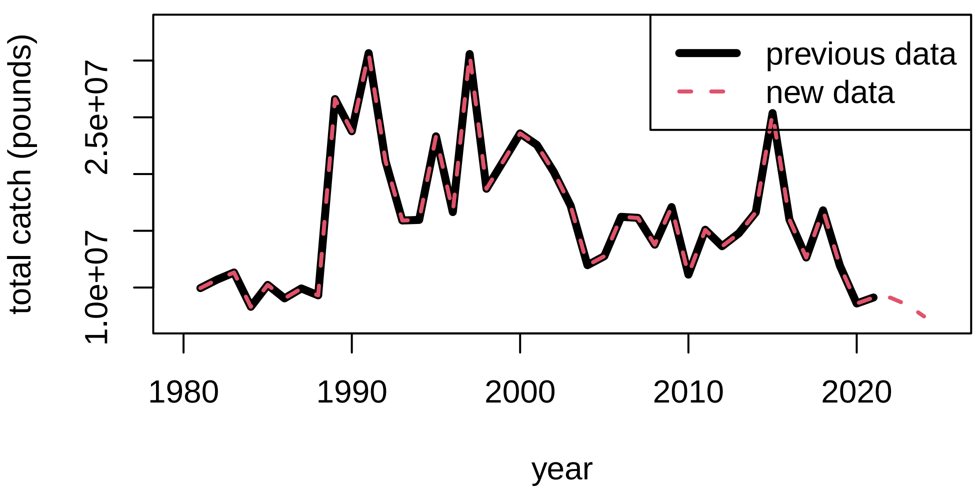
```
         CT    DE   FLE FLE/GA   FLW    GA    MA    MD    NC    NJ    NY    RI    SC    VA
  3       0     0     0      0   568     0     0     0     0     0     0     0     0     0
  4       0     0  1022      0     0     0     0     0     0     0     0     0     0     0
  5       0     0   312      0     0     0     0     0     0     0     0     0     0     0
  <NA>   14   114     0   1794     0    54    20   125  1344   108    73    62   644   165

       <NA>
  3       0
  4       0
  5       0
  <NA>    0
```

Now we can summarize the total dolphin landings in weight for the U.S. Atlantic, and compare to previous values sent for quota monitoring. We see that the values are almost identical.

```
# summarize by year and compare --------------------------

tot1 <- tapply(d1$LBSEST_SECWWT, d1$YEAR, sum, na.rm = T)
head(tot1)
```

```
    1981     1982     1983     1984     1985     1986
 9952644 10701659 11325575  8303073 10236339  9047438
```

```
old <- read.csv("data/RecreationalDolphinLandings_SAandGOM_MLarkin.csv", skip = 6)
plot(old$Year, old$ATL, type = "l", xlab = "year", ylab = "total catch (pounds)", lwd = 4,
     xlim = c(1980, 2025), ylim = c(7*10^6, 3.3*10^7))
lines(as.numeric(names(tot1)), tot1, lwd = 2, col = 2, lty = 2)
legend("topright", c("previous data", "new data"), col = c(1, 2), lwd = c(4, 2), lty = c(1,
```



2.3 Partition catches by area and sector

Now we will partition out the total catches according to the dolphin MSE areas. The areas are as follows:

- FLK: Florida Keys to Indian River County County (FL)
- NCFL: Brevard (FL) to Southern NC (South of Hatteras)
- NNC: Northern NC (North of Hatteras to NC/VA border)
- VBM: Virginia to Maine

```
# separate out 4 regions -------------------------

d1$region <- "VBM"

# select for FLK
# FL_REG codes: 4: SE FL- Miami-Dade to Indian River County.; 5: NE FL- Brevard to Nassau C
table(d1$FL_REG, d1$NEW_STA, useNA = "always")
```

```
        CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
  3      0    0    0      0  568    0    0    0    0    0    0    0    0    0
  4      0    0 1022      0    0    0    0    0    0    0    0    0    0    0
  5      0    0  312      0    0    0    0    0    0    0    0    0    0    0
  <NA>  14  114    0   1794    0   54   20  125 1344  108   73   62  644  165

      <NA>
  3      0
  4      0
  5      0
  <NA>   0
```

```
d1$region[which(d1$FL_REG == 3 | d1$FL_REG == 4)] <- "FLK"

# select for NCFL
d1$region[which(d1$FL_REG == 5)] <- "NCFL"
lis <- c("SC", "GA", "FLE/GA")           # NCFL states
d1$region[which(d1$NEW_STA %in% lis)] <- "NCFL"
# NC_REG codes: N: North of Cape Hatteras; S: South of Cape Hatteras
table(d1$NC_REG, d1$NEW_STA, useNA = "always")
```

```
         CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
         14  114 1334   1794  568   54   20  125  586  108   73   62  644  165
  N       0    0    0      0    0    0    0    0  420    0    0    0    0    0
  S       0    0    0      0    0    0    0    0  338    0    0    0    0    0
  <NA>    0    0    0      0    0    0    0    0    0    0    0    0    0    0

       <NA>
          0
  N       0
  S       0
  <NA>    0
```

```
d1$region[which(d1$NC == "S")] <- "NCFL"

# select for NNC
d1$region[which(d1$NC == "N")] <- "NNC"

table(d1$FL_REG, d1$NEW_STA, d1$region)
```

```
, ,  = FLK


      CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
  3    0    0    0      0  568    0    0    0    0    0    0    0    0    0
  4    0    0 1022      0    0    0    0    0    0    0    0    0    0    0
  5    0    0    0      0    0    0    0    0    0    0    0    0    0    0

, ,  = NCFL


      CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
  3    0    0    0      0    0    0    0    0    0    0    0    0    0    0
  4    0    0    0      0    0    0    0    0    0    0    0    0    0    0
  5    0    0  312      0    0    0    0    0    0    0    0    0    0    0

, ,  = NNC


      CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
```

```
  3    0    0    0      0    0    0    0    0    0    0    0    0    0    0
  4    0    0    0      0    0    0    0    0    0    0    0    0    0    0
  5    0    0    0      0    0    0    0    0    0    0    0    0    0    0

, ,  = VBM


     CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
  3    0    0    0      0    0    0    0    0    0    0    0    0    0    0
  4    0    0    0      0    0    0    0    0    0    0    0    0    0    0
  5    0    0    0      0    0    0    0    0    0    0    0    0    0    0
```

```
table(d1$NC_REG, d1$NEW_STA, d1$region)  # check classification
```

```
, ,  = FLK


     CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
       0    0 1022      0  568    0    0    0    0    0    0    0    0    0
  N    0    0    0      0    0    0    0    0    0    0    0    0    0    0
  S    0    0    0      0    0    0    0    0    0    0    0    0    0    0

, ,  = NCFL


     CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
       0    0  312   1794    0   54    0    0    0    0    0    0  644    0
  N    0    0    0      0    0    0    0    0    0    0    0    0    0    0
  S    0    0    0      0    0    0    0    0  338    0    0    0    0    0

, ,  = NNC


     CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
       0    0    0      0    0    0    0    0    0    0    0    0    0    0
  N    0    0    0      0    0    0    0    0  420    0    0    0    0    0
  S    0    0    0      0    0    0    0    0    0    0    0    0    0    0

, ,  = VBM
```

```
      CT  DE FLE FLE/GA FLW  GA  MA  MD  NC  NJ  NY  RI  SC  VA
      14 114   0      0   0   0  20 125 586 108  73  62   0 165
  N    0   0   0      0   0   0   0   0   0   0   0   0   0   0
  S    0   0   0      0   0   0   0   0   0   0   0   0   0   0
```

Now we will separate out the catches by recreational sector and plot the totals by region and sector.

```
# separate out by for hire vs. private rec --------------------------
d1$sector <- "ForHire"

d1$sector[which(d1$NEW_MODEN == "Shore")] <- "Rec"
d1$sector[which(d1$NEW_MODEN == "Priv")]  <- "Rec"

table(d1$sector, d1$NEW_MODEN, useNA = "always")    # check classification
```

```
           Cbt Cbt/Hbt  Hbt Priv Shore <NA>
  ForHire 1616     119 2912    0     0    0
  Rec        0       0    0 1750    22    0
  <NA>       0       0    0    0     0    0
```

```
tot <- tapply(d1$LBSEST_SECWWT, list(d1$YEAR, d1$region, d1$sector), sum, na.rm = T)

head(tot)
```

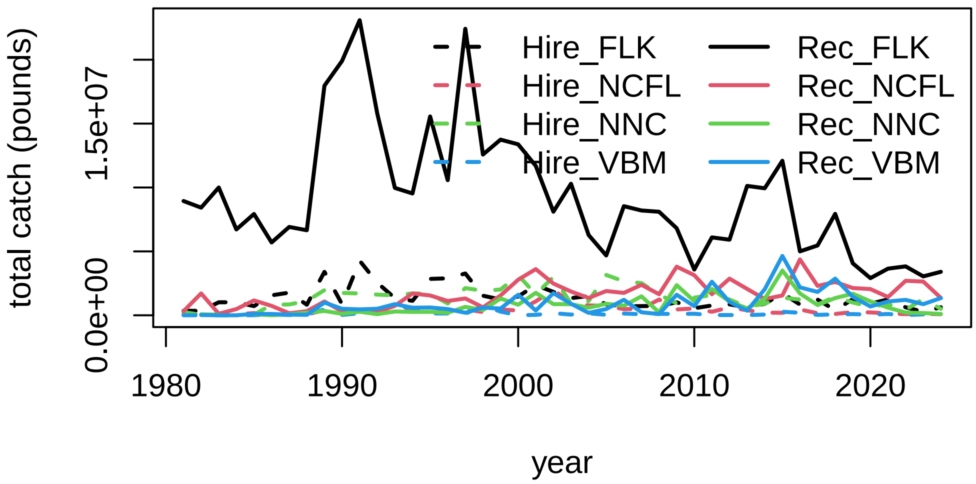
```
, , ForHire

           FLK      NCFL       NNC      VBM
1981  354639.5  74702.50 256755.24  1400.247
1982  347458.2  92271.42  27578.44  2749.224
1983 1020005.8  41784.04  88690.53  5379.177
1984 1021086.8  74658.81        NA  4203.048
1985  717523.0 172989.65  58514.13  8880.857
1986 1562354.0  63555.89 817192.26 61658.193

, , Rec
```

```
          FLK      NCFL       NNC        VBM
1981  8942641  322505.7        NA         NA
1982  8422673 1711834.7 78367.161  18727.290
1983 10000662  119370.4 44226.032   5456.237
1984  6717044  486079.7        NA      0.000
1985  7925301 1162708.4 77477.096 112944.274
1986  5699311  736872.0  2549.586 103945.134
```

```
fin <- data.frame(cbind(tot[, , 1], tot[, , 2]))
names(fin) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Re
head(fin)
```

```
      Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM  Rec_FLK  Rec_NCFL   Rec_NNC
1981  354639.5  74702.50 256755.24  1400.247  8942641  322505.7        NA
1982  347458.2  92271.42  27578.44  2749.224  8422673 1711834.7 78367.161
1983 1020005.8  41784.04  88690.53  5379.177 10000662  119370.4 44226.032
1984 1021086.8  74658.81        NA  4203.048  6717044  486079.7        NA
1985  717523.0 172989.65  58514.13  8880.857  7925301 1162708.4 77477.096
1986 1562354.0  63555.89 817192.26 61658.193  5699311  736872.0  2549.586
        Rec_VBM
1981         NA
1982  18727.290
1983   5456.237
1984      0.000
1985 112944.274
1986 103945.134
```

```
matplot(rownames(fin), fin, lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), type = "l", l
        xlab = "year", ylab = "total catch (pounds)")
legend("topright", colnames(fin), lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), lwd = 2
```




```
# check that totals match
cor(rowSums(fin, na.rm = T), tot1)
```

```
[1] 1
```

```
mean(rowSums(fin, na.rm = T) - tot1)   # very tiny differences
```

```
[1] 4.233284e-11
```

Here are the catches (in pounds) for years 1981 to 2024.

```
fin
```

```
      Hire_FLK  Hire_NCFL   Hire_NNC   Hire_VBM  Rec_FLK  Rec_NCFL    Rec_NNC
1981  354639.5   74702.50  256755.24   1400.247  8942641  322505.7         NA
1982  347458.2   92271.42   27578.44   2749.224  8422673 1711834.7  78367.161
1983 1020005.8   41784.04   88690.53   5379.177 10000662  119370.4  44226.032
1984 1021086.8   74658.81         NA   4203.048  6717044  486079.7         NA
1985  717523.0  172989.65   58514.13   8880.857  7925301 1162708.4  77477.096
```

```
1986 1562354.0   63555.89  817192.26  61658.193  5699311  736872.0    2549.586
1987 1774558.7   61243.58  848994.25  24967.877  6913585  161627.1   79540.365
1988  826480.6   62392.87 1114627.05  94676.212  6655923  321023.1  202768.950
1989 3398703.5  405140.01 1970776.23 451477.768 17965047 1079951.5  339371.567
1990  868572.6  277783.17 1749965.55  31079.442 19900865  312333.3  118539.960
1991 4268773.7  225552.64 1706751.50 159172.855 23093044  457667.1  277389.646
1992 2548407.4  189514.34 1619685.06 161149.703 15788853  245395.8   89160.860
1993 1343093.6  391254.33 1555850.09 741955.369  9967356  742741.8  296399.199
1994 1131831.7  437642.72 1713981.98 573551.716  9532763 1708942.1  264907.193
1995 2848691.7  667475.30 1673435.48 145798.587 15558489 1550136.0  271442.073
1996 2892436.3  309604.42  923128.81 141976.431 10574454 1121937.8  196579.537
1997 3271271.6  466891.67 2129471.46 110115.485 22423256 1316502.7  660257.843
1998 1532460.0  252620.73 1921834.08 779364.855 12578694  619824.1  412304.292
1999 1238462.0  457880.10 2014752.29 288625.130 13745135 1548217.4 1314805.374
2000 1471481.3  369553.33 3139555.98   3016.589 13379647 2784424.2  826980.337
2001 2361751.6 1090421.00 1638707.54  34335.241 11690170 3614370.3 1760721.144
2002 1828918.7 1993778.99 3005172.92 168406.810  8105262 2483946.6  871930.660
2003 1342339.1  958997.12  928894.01  55690.419 10294918 1860435.4  868200.254
2004 1509660.5  771820.65 1116373.11 155510.796  6281794 1354303.7  593257.716
2005  858090.6  734896.20 3153480.89  38020.553  4681210 1898945.9  904360.210
2006  700089.5  483337.12 2656240.22 136715.477  8540432 1751853.5  749541.413
2007  718651.9  506776.30 2518430.49  84908.300  8200488 2383417.1 1483630.590
2008  736213.2 1235299.71 1620865.87 113592.877  8100550 1648662.2  223565.185
2009 1016583.6  458562.82  928419.86 106262.915  6809447 3805127.7 2359177.817
2010  552080.9  534671.06 1384434.93 120050.410  3579591 3139199.6 1110479.730
2011  768597.7  275546.65 1577019.96  17844.729  6097050 1651096.4 2063315.642
2012  856261.4  615430.34 1236410.76  21887.318  5916144 2867599.1 1082607.063
2013  614150.1  394570.11  640465.60  16064.763 10124799 2064256.6  540676.961
2014  897508.4  210198.76  916695.96  56760.758  9940020 1272477.3 1289704.416
2015 1729825.9  196326.97 1374908.91 277867.986 12093928 1548428.5 3505194.743
2016  841662.5  454053.51 1236565.76 195837.854  5002273 4372622.6 1704661.669
2017 1227827.8  161188.31  835080.86  31185.061  5464893 2297930.0  801324.151
2018  401703.0  105328.49 1465087.58  72211.785  7935729 2606091.5 1347724.203
2019 1261827.1  250624.40  998102.45  93718.699  4064867 2132483.1 1687239.863
2020  910431.4  225493.79  667501.79  51621.152  2906913 2052222.0 1089245.171
2021 1244842.2  146893.15  871760.94 100997.979  3652596 1428508.3  596672.480
2022  627020.6   90504.71  427953.34  18514.133  3828113 2701619.8  211394.537
2023  274532.5  118528.80 1281254.55  72728.020  3045801 2639445.1  175189.224
2024  629327.2   85572.19  506558.07  18752.239  3404227 1358519.1  106307.278
        Rec_VBM
1981         NA
```

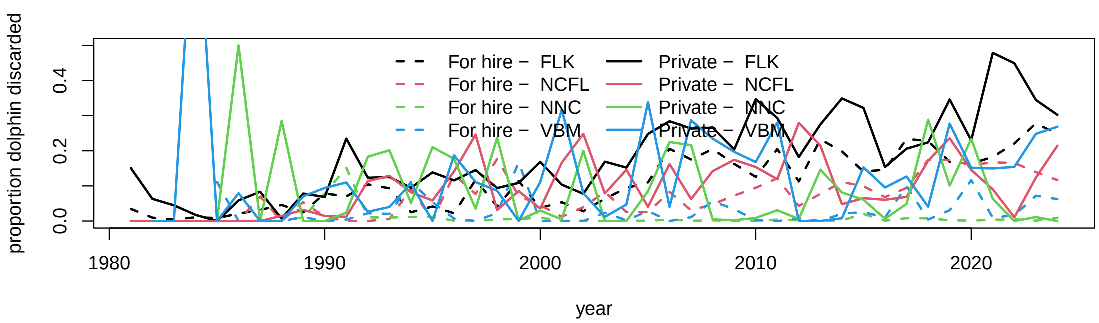
```
1982   18727.290
1983    5456.237
1984       0.000
1985  112944.274
1986  103945.134
1987   52139.772
1988   33566.163
1989  996945.390
1990  509424.981
1991  464786.733
1992  509297.261
1993  871948.601
1994  594467.414
1995  609302.619
1996  486106.187
1997  197871.963
1998  606769.298
1999  523623.688
2000 1608478.920
2001  372623.273
2002 1732356.036
2003  904780.558
2004  186538.567
2005  489236.451
2006 1214497.040
2007  236769.123
2008   96818.461
2009 1607919.004
2010  717410.858
2011 2626410.052
2012 1041694.282
2013  364492.286
2014 2040329.550
2015 4645796.249
2016 2186126.672
2017 1826822.849
2018 2871124.410
2019 1440420.940
2020  689099.471
2021 1083268.811
2022 1206278.324
```

```
2023  872490.886
2024 1346606.842
```

2.4 Looking at extent of discarding

Discard estimates (known as B2 estimates) are provided in numbers by the MRIP survey but are not available in the Southeast Region Headboat Survey. We can look at the overall numbers of dolphinfish caught versus reported released from the MRIP survey to get an understanding of the extent of discarding.

```
d1$perdis <- d1$B2 / (d1$AB1 + d1$B2)  # percent discarded is B2 (fish released alive) / to

# calculate mean discard rate by year, region, sector ----------------
dis <- tapply(d1$perdis, list(d1$YEAR, d1$region, d1$sector), mean, na.rm = T)
disr <- tapply(d1$perdis, list(d1$region, d1$sector), mean, na.rm = T)

matplot(rownames(dis), cbind(dis[, , 1], dis[, , 2]), lty = c(rep(2, 4), rep(1, 4)), col =
        type = "l", lwd = 2, xlab = "year", ylab = "proportion dolphin discarded", ylim = c
legend("top", c(paste("For hire - ", colnames(dis)), paste("Private - ", colnames(dis))),
       lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), lwd = 2, ncol = 2, bty = "n")
```




```
round(disr * 100, 2)  # discarding rates by region and sector
```

```
      ForHire   Rec
FLK     10.79 18.76
NCFL     9.44 11.23
```

```
NNC      0.94  9.62
VBM      2.94 12.69
```

According to the MRIP survey and the reported numbers, discarding generally increases over time, excluding some highly anomalous estimates from the early years which are driven by low sample sizes. Discarding is generally higher in the private sector compared to the for-hire sector, and is highest in the Florida Keys region.

2.5 Split the catches by season

The dolphin MSE operates on a seasonal time step and thus the catches and discards must by parsed apart by individual season and year. This is not straightforward because MRIP estimates are reported in two-month waves (Jan - Feb, Mar - Apr, May - Jun, etc.), and the seasons are 3-month periods (Dec - Feb, Mar - May, Jun - Aug, Sep - Nov). For the two waves that have to be split into seasons (e.g., December is lumped with January and February; June is lumped with July and August), we split the wave 50/50%, i.e, we make the assumption that half of the catch in that wave occurs in one season and the other half occurs in the next season.

```
# define seasons ----------------------------
table(d1$WAVE, useNA = "always")
```

```
   0    1    2    3    4    5    6 <NA> 
   5  407  718 1230 1439 1095  581  944 
```

```
d1$YRWAVE <- paste0(d1$YEAR, "_", d1$WAVE)

totw <- tapply(d1$LBSEST_SECWWT, list(d1$YRWAVE, d1$region, d1$sector), sum, na.rm = T)

f <- data.frame(cbind(totw[, , 1], totw[, , 2]))
names(f) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Rec_
head(f)
```

```
         Hire_FLK Hire_NCFL   Hire_NNC Hire_VBM   Rec_FLK  Rec_NCFL Rec_NNC
1981_2         NA        NA         NA       NA 2793285.5 270968.46      NA
1981_3 300873.764        NA 254049.420       NA 3915698.3        NA      NA
```

```
1981_4   50017.460        NA   2705.821        NA  905621.2  51537.26        NA
1981_5    3748.301        NA         NA        NA  479834.0        NA        NA
1981_6          NA        NA         NA        NA  848201.5        NA        NA
1981_NA         NA   74702.5         NA  1400.247        NA        NA        NA
        Rec_VBM
1981_2       NA
1981_3       NA
1981_4       NA
1981_5       NA
1981_6       NA
1981_NA      NA
```

```
f$YEAR <- as.numeric(substr(rownames(f), 1, 4))
f$WAVE <- as.numeric(substr(rownames(f), 6, 8))
head(f)   # table with catches separated by wave and year
```

```
          Hire_FLK Hire_NCFL   Hire_NNC Hire_VBM   Rec_FLK   Rec_NCFL Rec_NNC
1981_2          NA        NA         NA       NA 2793285.5 270968.46      NA
1981_3  300873.764        NA 254049.420       NA 3915698.3        NA      NA
1981_4   50017.460        NA   2705.821       NA  905621.2  51537.26      NA
1981_5    3748.301        NA         NA       NA  479834.0        NA      NA
1981_6          NA        NA         NA       NA  848201.5        NA      NA
1981_NA         NA   74702.5         NA 1400.247        NA        NA      NA
        Rec_VBM YEAR WAVE
1981_2       NA 1981    2
1981_3       NA 1981    3
1981_4       NA 1981    4
1981_5       NA 1981    5
1981_6       NA 1981    6
1981_NA      NA 1981   NA
```

```
f[is.na(f)] <- 0

f <- f[which(f$YEAR >= 1985), ]
f <- f[which(f$YEAR <= 2024), ]

table(f$YEAR)
```

```
1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000
   7    7    7    7    7    7    7    7    7    7    7    7    6    6    6    6
2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016
   6    6    6    6    6    6    6    6    6    6    6    6    7    7    7    6
2017 2018 2019 2020 2021 2022 2023 2024
   6    6    6    6    6    6    6    6
```

```r
table(f$WAVE)
```

```
 0  1  2  3  4  5  6
15 40 40 40 40 40 40
```

```r
# deal with NAs in many locations ---------------
# calculate percentages of catch by wave, for each region and fleet
avw <- tapply(d1$LBSEST_SECWWT, list(d1$WAVE, d1$region, d1$sector), sum, na.rm = T)
avw <- avw[-1, , ]
avw <- data.frame(cbind(avw[, , 1], avw[, , 2]))
names(avw) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Re
avwp <- apply(avw, 2, function (x) x / sum(x, na.rm = T))
#colSums(avwp)

# go through each column, find NAs and fill in waves by percentages
pre_colsum <- colSums(f, na.rm = T)  # column sums to check later

lis <- which(f$WAVE == 0)  # list of rows with NA wave data

for (i in 1:8) {        # loop through each column
  for(j in lis) {       # loop through list of NA waves
    if (f[j, i] > 0) {  # if there are NA wave catches, split by known proportion from othe
      miscatch <- avwp[, i] * f[j, i]   # missing catch is known proportion * NA wave catch
      f[which(f$YEAR == f$YEAR[j] & f$WAVE != 0), i] <- f[which(f$YEAR == f$YEAR[j] & f$WAV
      f[j, i] <- 0   # set to zero after catch is attributed to waves
    }
  }
}
f[lis, 1:8] # should be all zeros now
```

```
        Hire_FLK Hire_NCFL Hire_NNC Hire_VBM Rec_FLK Rec_NCFL Rec_NNC Rec_VBM
```

```
1985_NA         0          0         0         0        0         0        0        0
1986_NA         0          0         0         0        0         0        0        0
1987_NA         0          0         0         0        0         0        0        0
1988_NA         0          0         0         0        0         0        0        0
1989_NA         0          0         0         0        0         0        0        0
1990_NA         0          0         0         0        0         0        0        0
1991_NA         0          0         0         0        0         0        0        0
1992_NA         0          0         0         0        0         0        0        0
1993_NA         0          0         0         0        0         0        0        0
1994_NA         0          0         0         0        0         0        0        0
1995_NA         0          0         0         0        0         0        0        0
1996_NA         0          0         0         0        0         0        0        0
2013_0          0          0         0         0        0         0        0        0
2014_0          0          0         0         0        0         0        0        0
2015_0          0          0         0         0        0         0        0        0
```

```
pos_colsum <- colSums(f, na.rm = T)
pre_colsum[1:8] == pos_colsum[1:8]  # check that catch was parsed correctly - column sums s
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM
     TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE
```

```
f <- f[which(f$WAVE != 0), ]  # remove NA waves which now contain no catch

# split waves 3 and 6 50/50% to parse into seasons ------------------
lis <- which(f$WAVE == 3 | f$WAVE == 6)  # list of 3 and 6 waves for all years

for (i in lis) {           # go through list of waves
  temp <- f[i, ] / 2       # split catch in half
  temp[9] <- f$YEAR[i]     # reformat year and wave data columns
  temp[10] <- f$WAVE[i]
  temp <- rbind(temp, temp)   # create two half-waves and relabel with new wave reference n
  temp[1, 10] <- temp[1, 10] - 0.5
  temp[2, 10] <- temp[2, 10] + 0.5
  f <- rbind(f, temp)         # concatenate with main data table
  }

f <- f[-lis, ]  # remove waves that have been split in half
```

```r
pos_colsum <- colSums(f, na.rm = T)
pre_colsum[1:8] == pos_colsum[1:8]  # check that catch was parsed correctly - column sums s
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM
     TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE
```

```r
f <- f[order(f$YEAR, f$WAVE), ]  # reorder table in chronological order

f$YEAR[which(f$WAVE == 6.5)] <- f$YEAR[which(f$WAVE == 6.5)] + 1  # add year to December wa
f$WAVE[which(f$WAVE == 6.5)] <- 0.5  # change December wave to small number so it gets summ

table(f$WAVE)
```

```
0.5   1   2 2.5 3.5   4   5 5.5
 40  40  40  40  40  40  40  40
```

```r
f$seas <- cut(f$WAVE, breaks = c(0, 1.5, 3, 4.5, 6))
#table(f$seas)
f$seas1 <- NA
f$seas1[which(f$seas == "(0,1.5]")] <- "DJF"  # Dec Jan Feb is 0.5 and 1
f$seas1[which(f$seas == "(1.5,3]")] <- "MAM"  # Mar Apr May is 2 and 2.5
f$seas1[which(f$seas == "(3,4.5]")] <- "JJA"  # Jun Jul Aug is 3.5 and 4
f$seas1[which(f$seas == "(4.5,6]")] <- "SON"  # Sep Oct Nov is 5 and 5.5

temp <- tapply(f[, 1], list(f$seas1, f$YEAR), sum, na.rm = T)
temp1 <- matrix(temp)
for (i in 2:8) {
  temp2 <- matrix(tapply(f[, i], list(f$seas1, f$YEAR), sum, na.rm = T))
  temp1 <- cbind(temp1, temp2)
  }

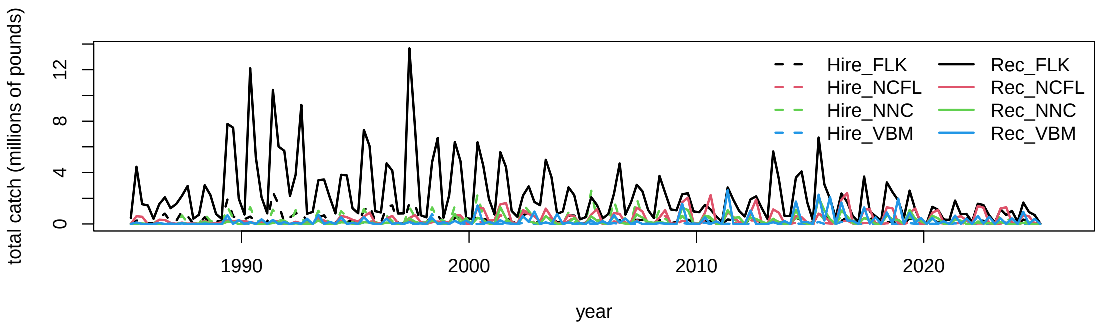
findat <- data.frame(sort(rep(as.numeric(colnames(temp)), 4)),
                     rep(rownames(temp), ncol(temp)), temp1)
names(findat) <- c("year", "season", names(f)[1:8])

colSums(f[, 1:8], na.rm = T) == colSums(findat[, 3:10], na.rm = T)  # check that total catc
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM
     TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE
```

We plot the results by region and sector, over the time series of year and season. We can clearly see the seasonality in the catch by region.

```
findat$yrseas <- findat$year + as.numeric(as.factor(findat$season))/4-0.125
matplot(findat$yrseas, findat[, 3:10]/10^6, lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2
        xlab = "year", ylab = "total catch (millions of pounds)")
legend("topright", colnames(findat[, 3:10]), lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4,
```




```
# check that totals match
cbind(colSums(findat[, 3:10], na.rm = T), colSums(fin[-c(1:5), ], na.rm = T))
```

```
               [,1]      [,2]
Hire_FLK   55604999  54887476
Hire_NCFL  16909916  16736926
Hire_NNC   57944903  57886388
Hire_VBM    5836953   5828072
Rec_FLK   371462933 363537632
Rec_NCFL   70803866  69641157
Rec_NNC    32556596  32479119
Rec_VBM    40081529  39968585
```

```
# the values should be close but slightly >1 because the last data set includes December 19
colSums(findat[, 3:10], na.rm = T) / colSums(fin[-c(1:5), ], na.rm = T)
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM
 1.013073  1.010336  1.001011  1.001524  1.021800  1.016696  1.002385  1.002826
```

```
write.csv(findat, file = "data/MRIP_catch_by_season_4_areas.csv", row.names = FALSE)
```

Here we will format the data for input to the operating model

```
tab3 <- findat

flt <- c(rep("ForHire", nrow(tab3)*4), rep("Rec", nrow(tab3)*4))

yrmon <- as.numeric(tab3$yrseas)
yrs <- floor(yrmon)
qrt <- (yrmon - floor(yrmon)) * 4 + 0.5
area <- c(rep("FLK", nrow(tab3)), rep("NCFL", nrow(tab3)),
          rep("NNC", nrow(tab3)), rep("VBM", nrow(tab3)))
area <- rep(area, 2)


findat1 <- data.frame(yrs, qrt, flt, area, matrix(as.matrix(tab3[,3:10])))
findat1
```

```
      yrs qrt     flt area matrix.as.matrix.tab3...3.10...
1     1985   1 ForHire  FLK                    4.462090e+03
2     1985   2 ForHire  FLK                    2.534874e+05
3     1985   3 ForHire  FLK                    3.808114e+05
4     1985   4 ForHire  FLK                    4.146999e+04
5     1986   1 ForHire  FLK                    4.054541e+04
6     1986   2 ForHire  FLK                    5.011846e+05
7     1986   3 ForHire  FLK                    8.024703e+05
8     1986   4 ForHire  FLK                    1.511596e+05
9     1987   1 ForHire  FLK                    2.414829e+05
10    1987   2 ForHire  FLK                    9.787720e+05
11    1987   3 ForHire  FLK                    6.087522e+05
12    1987   4 ForHire  FLK                    2.491889e+04
13    1988   1 ForHire  FLK                    5.112349e+04
14    1988   2 ForHire  FLK                    5.236336e+05
15    1988   3 ForHire  FLK                    2.796967e+04
16    1988   4 ForHire  FLK                    1.960574e+05
17    1989   1 ForHire  FLK                    2.524921e+05
```

```
18   1989   2 ForHire  FLK                                1.898385e+06
19   1989   3 ForHire  FLK                                5.918365e+05
20   1989   4 ForHire  FLK                                3.962056e+05
21   1990   1 ForHire  FLK                                3.673921e+05
22   1990   2 ForHire  FLK                                5.735168e+05
23   1990   3 ForHire  FLK                                1.672854e+05
24   1990   4 ForHire  FLK                                5.481602e+04
25   1991   1 ForHire  FLK                                1.200051e+05
26   1991   2 ForHire  FLK                                2.463709e+06
27   1991   3 ForHire  FLK                                1.530429e+06
28   1991   4 ForHire  FLK                                1.329583e+05
29   1992   1 ForHire  FLK                                5.113129e+05
30   1992   2 ForHire  FLK                                8.083241e+05
31   1992   3 ForHire  FLK                                9.688738e+05
32   1992   4 ForHire  FLK                                1.566697e+05
33   1993   1 ForHire  FLK                                2.333242e+05
34   1993   2 ForHire  FLK                                5.197939e+05
35   1993   3 ForHire  FLK                                6.692056e+05
36   1993   4 ForHire  FLK                                3.734484e+04
37   1994   1 ForHire  FLK                                8.981145e+04
38   1994   2 ForHire  FLK                                5.934543e+05
39   1994   3 ForHire  FLK                                1.406221e+05
40   1994   4 ForHire  FLK                                2.984649e+05
41   1995   1 ForHire  FLK                                1.390055e+05
42   1995   2 ForHire  FLK                                1.039095e+06
43   1995   3 ForHire  FLK                                1.461029e+06
44   1995   4 ForHire  FLK                                2.027951e+05
45   1996   1 ForHire  FLK                                7.773956e+04
46   1996   2 ForHire  FLK                                1.352282e+06
47   1996   3 ForHire  FLK                                1.439914e+06
48   1996   4 ForHire  FLK                                4.336462e+04
49   1997   1 ForHire  FLK                                3.874707e+04
50   1997   2 ForHire  FLK                                1.629481e+06
51   1997   3 ForHire  FLK                                1.514418e+06
52   1997   4 ForHire  FLK                                8.428625e+04
53   1998   1 ForHire  FLK                                9.909438e+04
54   1998   2 ForHire  FLK                                6.713363e+05
55   1998   3 ForHire  FLK                                6.563503e+05
56   1998   4 ForHire  FLK                                9.008397e+04
57   1999   1 ForHire  FLK                                1.387479e+05
58   1999   2 ForHire  FLK                                4.497292e+05
```

59	1999	3	ForHire	FLK	5.063335e+05
60	1999	4	ForHire	FLK	1.268359e+05
61	2000	1	ForHire	FLK	8.315590e+04
62	2000	2	ForHire	FLK	7.265192e+05
63	2000	3	ForHire	FLK	4.308598e+05
64	2000	4	ForHire	FLK	2.502685e+05
65	2001	1	ForHire	FLK	2.460745e+05
66	2001	2	ForHire	FLK	1.233226e+06
67	2001	3	ForHire	FLK	7.170208e+05
68	2001	4	ForHire	FLK	1.353495e+05
69	2002	1	ForHire	FLK	1.250031e+05
70	2002	2	ForHire	FLK	7.042666e+05
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72	2002	4	ForHire	FLK	2.144944e+05
73	2003	1	ForHire	FLK	1.173219e+05
74	2003	2	ForHire	FLK	6.037893e+05
75	2003	3	ForHire	FLK	4.592291e+05
76	2003	4	ForHire	FLK	1.306186e+05
77	2004	1	ForHire	FLK	1.665554e+05
78	2004	2	ForHire	FLK	6.537249e+05
79	2004	3	ForHire	FLK	6.539262e+05
80	2004	4	ForHire	FLK	6.991629e+04
81	2005	1	ForHire	FLK	8.617154e+04
82	2005	2	ForHire	FLK	3.823569e+05
83	2005	3	ForHire	FLK	3.634808e+05
84	2005	4	ForHire	FLK	4.281886e+04
85	2006	1	ForHire	FLK	6.246935e+04
86	2006	2	ForHire	FLK	2.945357e+05
87	2006	3	ForHire	FLK	2.947750e+05
88	2006	4	ForHire	FLK	4.741635e+04
89	2007	1	ForHire	FLK	3.433560e+04
90	2007	2	ForHire	FLK	3.291462e+05
91	2007	3	ForHire	FLK	3.047614e+05
92	2007	4	ForHire	FLK	4.854292e+04
93	2008	1	ForHire	FLK	8.213770e+04
94	2008	2	ForHire	FLK	3.145215e+05
95	2008	3	ForHire	FLK	2.804606e+05
96	2008	4	ForHire	FLK	4.715174e+04
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99	2009	3	ForHire	FLK	3.926185e+05

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137  2019   1 ForHire  FLK                      2.832672e+04
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140  2019   4 ForHire  FLK                      4.160198e+05
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149	2022	1	ForHire	FLK	2.639086e+04
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161	2025	1	ForHire	FLK	1.563611e+04
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174	1987	2	ForHire	NCFL	3.073598e+04
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444	2013	4	ForHire	NNC	1.719227e+05
445	2014	1	ForHire	NNC	6.398102e+02
446	2014	2	ForHire	NNC	4.865816e+05
447	2014	3	ForHire	NNC	3.421213e+05
448	2014	4	ForHire	NNC	8.761344e+04
449	2015	1	ForHire	NNC	3.796263e+02
450	2015	2	ForHire	NNC	8.873625e+05
451	2015	3	ForHire	NNC	3.237039e+05
452	2015	4	ForHire	NNC	1.537656e+05
453	2016	1	ForHire	NNC	1.007698e+04
454	2016	2	ForHire	NNC	6.548437e+05
455	2016	3	ForHire	NNC	3.086638e+05
456	2016	4	ForHire	NNC	2.719797e+05
457	2017	1	ForHire	NNC	1.078562e+03
458	2017	2	ForHire	NNC	4.489343e+05
459	2017	3	ForHire	NNC	2.993657e+05
460	2017	4	ForHire	NNC	8.379875e+04
461	2018	1	ForHire	NNC	2.982117e+03
462	2018	2	ForHire	NNC	7.496388e+05
463	2018	3	ForHire	NNC	3.938811e+05
464	2018	4	ForHire	NNC	3.213315e+05
465	2019	1	ForHire	NNC	2.361142e+02
466	2019	2	ForHire	NNC	5.174288e+05
467	2019	3	ForHire	NNC	3.287088e+05
468	2019	4	ForHire	NNC	1.493810e+05

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470  2020   2 ForHire  NNC                        3.859802e+05
471  2020   3 ForHire  NNC                        1.904235e+05
472  2020   4 ForHire  NNC                        9.085066e+04
473  2021   1 ForHire  NNC                        2.474158e+02
474  2021   2 ForHire  NNC                        4.377578e+05
475  2021   3 ForHire  NNC                        1.903206e+05
476  2021   4 ForHire  NNC                        2.433248e+05
477  2022   1 ForHire  NNC                        3.577490e+02
478  2022   2 ForHire  NNC                        1.877862e+05
479  2022   3 ForHire  NNC                        1.499484e+05
480  2022   4 ForHire  NNC                        8.883334e+04
481  2023   1 ForHire  NNC                        1.385437e+03
482  2023   2 ForHire  NNC                        6.917578e+05
483  2023   3 ForHire  NNC                        4.855651e+05
484  2023   4 ForHire  NNC                        1.039317e+05
485  2024   1 ForHire  NNC                        0.000000e+00
486  2024   2 ForHire  NNC                        2.081499e+05
487  2024   3 ForHire  NNC                        1.385556e+05
488  2024   4 ForHire  NNC                        1.597430e+05
489  2025   1 ForHire  NNC                        1.094585e+02
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491  2025   3 ForHire  NNC                                  NA
492  2025   4 ForHire  NNC                                  NA
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497  1986   1 ForHire  VBM                        1.827329e-01
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501  1987   1 ForHire  VBM                        1.433616e-01
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503  1987   3 ForHire  VBM                        1.014313e+02
504  1987   4 ForHire  VBM                        1.782641e+04
505  1988   1 ForHire  VBM                        1.946561e-01
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508  1988   4 ForHire  VBM                        8.544770e+04
509  1989   1 ForHire  VBM                        1.902211e-01
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516	1990	4	ForHire	VBM	9.387198e+03
517	1991	1	ForHire	VBM	6.994376e-01
518	1991	2	ForHire	VBM	9.711731e+04
519	1991	3	ForHire	VBM	6.804028e+02
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521	1992	1	ForHire	VBM	1.305756e+00
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523	1992	3	ForHire	VBM	2.967570e+02
524	1992	4	ForHire	VBM	2.973731e+04
525	1993	1	ForHire	VBM	5.695040e-01
526	1993	2	ForHire	VBM	7.241465e+05
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529	1994	1	ForHire	VBM	1.075557e+00
530	1994	2	ForHire	VBM	5.608290e+05
531	1994	3	ForHire	VBM	4.745844e+03
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534	1995	2	ForHire	VBM	1.151956e+05
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536	1995	4	ForHire	VBM	2.880463e+04
537	1996	1	ForHire	VBM	4.132299e+01
538	1996	2	ForHire	VBM	4.977416e+04
539	1996	3	ForHire	VBM	9.135979e+02
540	1996	4	ForHire	VBM	9.128843e+04
541	1997	1	ForHire	VBM	2.450390e-01
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543	1997	3	ForHire	VBM	1.425417e+03
544	1997	4	ForHire	VBM	9.832395e+04
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547	1998	3	ForHire	VBM	1.270259e+03
548	1998	4	ForHire	VBM	2.276001e+04
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554	2000	2	ForHire	VBM	1.590883e+03
555	2000	3	ForHire	VBM	4.318811e+02
556	2000	4	ForHire	VBM	9.839474e+02
557	2001	1	ForHire	VBM	9.876608e+00
558	2001	2	ForHire	VBM	1.849069e+04
559	2001	3	ForHire	VBM	3.416471e+03
560	2001	4	ForHire	VBM	1.241709e+04
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562	2002	2	ForHire	VBM	8.680582e+04
563	2002	3	ForHire	VBM	6.614244e+04
564	2002	4	ForHire	VBM	1.539632e+04
565	2003	1	ForHire	VBM	6.222483e+01
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571	2004	3	ForHire	VBM	1.585921e+03
572	2004	4	ForHire	VBM	3.854991e+04
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575	2005	3	ForHire	VBM	3.255974e+02
576	2005	4	ForHire	VBM	9.044386e+03
577	2006	1	ForHire	VBM	0.000000e+00
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579	2006	3	ForHire	VBM	2.770928e+03
580	2006	4	ForHire	VBM	6.486364e+04
581	2007	1	ForHire	VBM	1.455036e+00
582	2007	2	ForHire	VBM	2.436482e+04
583	2007	3	ForHire	VBM	2.983204e+03
584	2007	4	ForHire	VBM	5.754550e+04
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586	2008	2	ForHire	VBM	9.670219e+04
587	2008	3	ForHire	VBM	8.914606e+02
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591	2009	3	ForHire	VBM	1.413382e+04

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605  2013   1 ForHire  VBM                     0.000000e+00
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608  2013   4 ForHire  VBM                     4.825515e+03
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624  2017   4 ForHire  VBM                     2.533937e+04
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626  2018   2 ForHire  VBM                     3.341788e+04
627  2018   3 ForHire  VBM                     2.440593e+03
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637	2021	1	ForHire	VBM	9.104998e+00
638	2021	2	ForHire	VBM	4.958988e+04
639	2021	3	ForHire	VBM	5.388107e+02
640	2021	4	ForHire	VBM	5.086524e+04
641	2022	1	ForHire	VBM	4.050165e+00
642	2022	2	ForHire	VBM	1.172606e+04
643	2022	3	ForHire	VBM	1.064174e+03
644	2022	4	ForHire	VBM	5.723903e+03
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648	2023	4	ForHire	VBM	3.913939e+04
649	2024	1	ForHire	VBM	0.000000e+00
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652	2024	4	ForHire	VBM	7.943312e+03
653	2025	1	ForHire	VBM	0.000000e+00
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655	2025	3	ForHire	VBM	NA
656	2025	4	ForHire	VBM	NA
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661	1986	1	Rec	FLK	4.020906e+05
662	1986	2	Rec	FLK	1.537010e+06
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665	1987	1	Rec	FLK	1.556820e+06
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683	1991	3	Rec	FLK	6.015731e+06
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804	2021	4	Rec	FLK	7.848983e+05
805	2022	1	Rec	FLK	1.633176e+05
806	2022	2	Rec	FLK	1.574723e+06
807	2022	3	Rec	FLK	1.470855e+06
808	2022	4	Rec	FLK	6.657709e+05
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812	2023	4	Rec	FLK	1.022943e+06
813	2024	1	Rec	FLK	1.741675e+05
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815	2024	3	Rec	FLK	9.281034e+05
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817	2025	1	Rec	FLK	3.138698e+04
818	2025	2	Rec	FLK	NA
819	2025	3	Rec	FLK	NA
820	2025	4	Rec	FLK	NA
821	1985	1	Rec	NCFL	0.000000e+00
822	1985	2	Rec	NCFL	6.036182e+05
823	1985	3	Rec	NCFL	5.590902e+05
824	1985	4	Rec	NCFL	0.000000e+00
825	1986	1	Rec	NCFL	0.000000e+00
826	1986	2	Rec	NCFL	3.338597e+05
827	1986	3	Rec	NCFL	3.017922e+05
828	1986	4	Rec	NCFL	8.768057e+04
829	1987	1	Rec	NCFL	1.353952e+04
830	1987	2	Rec	NCFL	1.077096e+05
831	1987	3	Rec	NCFL	1.505825e+04
832	1987	4	Rec	NCFL	3.885919e+04
833	1988	1	Rec	NCFL	0.000000e+00
834	1988	2	Rec	NCFL	2.502125e+05
835	1988	3	Rec	NCFL	2.737798e+04
836	1988	4	Rec	NCFL	4.343268e+04
837	1989	1	Rec	NCFL	1.399500e+05

838	1989	2	Rec NCFL	2.442929e+05
839	1989	3	Rec NCFL	2.253837e+05
840	1989	4	Rec NCFL	2.617131e+05
841	1990	1	Rec NCFL	2.086118e+05
842	1990	2	Rec NCFL	1.494802e+05
843	1990	3	Rec NCFL	4.926664e+04
844	1990	4	Rec NCFL	1.135865e+05
845	1991	1	Rec NCFL	0.000000e+00
846	1991	2	Rec NCFL	2.595592e+05
847	1991	3	Rec NCFL	1.863238e+05
848	1991	4	Rec NCFL	1.178413e+04
849	1992	1	Rec NCFL	0.000000e+00
850	1992	2	Rec NCFL	1.727708e+05
851	1992	3	Rec NCFL	4.607402e+04
852	1992	4	Rec NCFL	2.655088e+04
853	1993	1	Rec NCFL	0.000000e+00
854	1993	2	Rec NCFL	4.355714e+05
855	1993	3	Rec NCFL	2.752532e+05
856	1993	4	Rec NCFL	2.290328e+04
857	1994	1	Rec NCFL	3.723780e+04
858	1994	2	Rec NCFL	6.328460e+05
859	1994	3	Rec NCFL	5.279053e+05
860	1994	4	Rec NCFL	3.281162e+05
861	1995	1	Rec NCFL	1.918508e+05
862	1995	2	Rec NCFL	5.730691e+05
863	1995	3	Rec NCFL	8.825788e+05
864	1995	4	Rec NCFL	6.613425e+04
865	1996	1	Rec NCFL	2.835385e+04
866	1996	2	Rec NCFL	4.246640e+05
867	1996	3	Rec NCFL	6.274736e+05
868	1996	4	Rec NCFL	6.980021e+04
869	1997	1	Rec NCFL	0.000000e+00
870	1997	2	Rec NCFL	4.617975e+05
871	1997	3	Rec NCFL	6.912068e+05
872	1997	4	Rec NCFL	1.634984e+05
873	1998	1	Rec NCFL	0.000000e+00
874	1998	2	Rec NCFL	1.689554e+05
875	1998	3	Rec NCFL	1.873414e+05
876	1998	4	Rec NCFL	1.458389e+05
877	1999	1	Rec NCFL	1.459837e+05
878	1999	2	Rec NCFL	7.204270e+05

879	1999	3	Rec NCFL	6.613227e+05
880	1999	4	Rec NCFL	1.147492e+05
881	2000	1	Rec NCFL	2.342309e+04
882	2000	2	Rec NCFL	1.292909e+06
883	2000	3	Rec NCFL	1.223197e+06
884	2000	4	Rec NCFL	2.444075e+05
885	2001	1	Rec NCFL	3.043074e+05
886	2001	2	Rec NCFL	1.526796e+06
887	2001	3	Rec NCFL	1.626879e+06
888	2001	4	Rec NCFL	1.400650e+05
889	2002	1	Rec NCFL	4.023345e+04
890	2002	2	Rec NCFL	9.589750e+05
891	2002	3	Rec NCFL	7.704207e+05
892	2002	4	Rec NCFL	5.947019e+05
893	2003	1	Rec NCFL	1.996363e+05
894	2003	2	Rec NCFL	1.196275e+06
895	2003	3	Rec NCFL	5.198576e+05
896	2003	4	Rec NCFL	5.640292e+04
897	2004	1	Rec NCFL	5.587967e+04
898	2004	2	Rec NCFL	6.996628e+05
899	2004	3	Rec NCFL	5.903592e+05
900	2004	4	Rec NCFL	3.085426e+04
901	2005	1	Rec NCFL	2.566053e+04
902	2005	2	Rec NCFL	7.310031e+05
903	2005	3	Rec NCFL	1.137113e+06
904	2005	4	Rec NCFL	2.547820e+04
905	2006	1	Rec NCFL	5.351815e+03
906	2006	2	Rec NCFL	8.487273e+05
907	2006	3	Rec NCFL	7.853118e+05
908	2006	4	Rec NCFL	7.268633e+04
909	2007	1	Rec NCFL	1.582543e+05
910	2007	2	Rec NCFL	1.269099e+06
911	2007	3	Rec NCFL	9.717851e+05
912	2007	4	Rec NCFL	2.940661e+04
913	2008	1	Rec NCFL	2.669028e+04
914	2008	2	Rec NCFL	5.067091e+05
915	2008	3	Rec NCFL	1.070780e+06
916	2008	4	Rec NCFL	2.970883e+04
917	2009	1	Rec NCFL	2.401520e+04
918	2009	2	Rec NCFL	1.689311e+06
919	2009	3	Rec NCFL	2.046079e+06

920	2009	4	Rec NCFL	3.594953e+04
921	2010	1	Rec NCFL	2.454607e+04
922	2010	2	Rec NCFL	8.796385e+05
923	2010	3	Rec NCFL	2.247155e+06
924	2010	4	Rec NCFL	1.240658e+04
925	2011	1	Rec NCFL	3.455564e+04
926	2011	2	Rec NCFL	9.178266e+05
927	2011	3	Rec NCFL	5.412456e+05
928	2011	4	Rec NCFL	1.574686e+05
929	2012	1	Rec NCFL	0.000000e+00
930	2012	2	Rec NCFL	8.746241e+05
931	2012	3	Rec NCFL	1.894418e+06
932	2012	4	Rec NCFL	9.855716e+04
933	2013	1	Rec NCFL	2.144051e+04
934	2013	2	Rec NCFL	1.133952e+06
935	2013	3	Rec NCFL	8.739791e+05
936	2013	4	Rec NCFL	3.488479e+04
937	2014	1	Rec NCFL	0.000000e+00
938	2014	2	Rec NCFL	6.355006e+05
939	2014	3	Rec NCFL	5.279255e+05
940	2014	4	Rec NCFL	1.063765e+05
941	2015	1	Rec NCFL	2.674703e+03
942	2015	2	Rec NCFL	8.061179e+05
943	2015	3	Rec NCFL	5.417106e+05
944	2015	4	Rec NCFL	2.005999e+05
945	2016	1	Rec NCFL	3.551940e+04
946	2016	2	Rec NCFL	1.880964e+06
947	2016	3	Rec NCFL	2.411574e+06
948	2016	4	Rec NCFL	3.674186e+04
949	2017	1	Rec NCFL	2.128542e+04
950	2017	2	Rec NCFL	1.121239e+06
951	2017	3	Rec NCFL	1.145773e+06
952	2017	4	Rec NCFL	1.479721e+04
953	2018	1	Rec NCFL	4.798984e+04
954	2018	2	Rec NCFL	1.292812e+06
955	2018	3	Rec NCFL	1.214296e+06
956	2018	4	Rec NCFL	5.313994e+04
957	2019	1	Rec NCFL	1.805194e+04
958	2019	2	Rec NCFL	8.596577e+05
959	2019	3	Rec NCFL	1.202456e+06
960	2019	4	Rec NCFL	4.788760e+04

961	2020	1	Rec	NCFL	4.941575e+03
962	2020	2	Rec	NCFL	8.708714e+05
963	2020	3	Rec	NCFL	1.093047e+06
964	2020	4	Rec	NCFL	7.473885e+04
965	2021	1	Rec	NCFL	1.356446e+04
966	2021	2	Rec	NCFL	4.814084e+05
967	2021	3	Rec	NCFL	6.071702e+05
968	2021	4	Rec	NCFL	2.963646e+05
969	2022	1	Rec	NCFL	4.356505e+04
970	2022	2	Rec	NCFL	1.398654e+06
971	2022	3	Rec	NCFL	1.267020e+06
972	2022	4	Rec	NCFL	3.594554e+04
973	2023	1	Rec	NCFL	2.716461e+04
974	2023	2	Rec	NCFL	1.207814e+06
975	2023	3	Rec	NCFL	1.299211e+06
976	2023	4	Rec	NCFL	1.052561e+05
977	2024	1	Rec	NCFL	0.000000e+00
978	2024	2	Rec	NCFL	6.938219e+05
979	2024	3	Rec	NCFL	5.986244e+05
980	2024	4	Rec	NCFL	3.303640e+04
981	2025	1	Rec	NCFL	3.303640e+04
982	2025	2	Rec	NCFL	NA
983	2025	3	Rec	NCFL	NA
984	2025	4	Rec	NCFL	NA
985	1985	1	Rec	NNC	0.000000e+00
986	1985	2	Rec	NNC	3.873855e+04
987	1985	3	Rec	NNC	3.873855e+04
988	1985	4	Rec	NNC	0.000000e+00
989	1986	1	Rec	NNC	0.000000e+00
990	1986	2	Rec	NNC	2.549586e+03
991	1986	3	Rec	NNC	0.000000e+00
992	1986	4	Rec	NNC	0.000000e+00
993	1987	1	Rec	NNC	0.000000e+00
994	1987	2	Rec	NNC	7.412236e+04
995	1987	3	Rec	NNC	0.000000e+00
996	1987	4	Rec	NNC	5.418005e+03
997	1988	1	Rec	NNC	0.000000e+00
998	1988	2	Rec	NNC	1.909134e+05
999	1988	3	Rec	NNC	1.185557e+04
1000	1988	4	Rec	NNC	0.000000e+00
1001	1989	1	Rec	NNC	0.000000e+00

1002	1989	2	Rec	NNC	1.902856e+05
1003	1989	3	Rec	NNC	1.465598e+05
1004	1989	4	Rec	NNC	1.588620e+03
1005	1990	1	Rec	NNC	9.376058e+02
1006	1990	2	Rec	NNC	1.017529e+05
1007	1990	3	Rec	NNC	1.379817e+04
1008	1990	4	Rec	NNC	2.988903e+03
1009	1991	1	Rec	NNC	0.000000e+00
1010	1991	2	Rec	NNC	7.471727e+04
1011	1991	3	Rec	NNC	1.882011e+05
1012	1991	4	Rec	NNC	1.447126e+04
1013	1992	1	Rec	NNC	0.000000e+00
1014	1992	2	Rec	NNC	4.499024e+04
1015	1992	3	Rec	NNC	2.068819e+04
1016	1992	4	Rec	NNC	2.348243e+04
1017	1993	1	Rec	NNC	0.000000e+00
1018	1993	2	Rec	NNC	2.759011e+05
1019	1993	3	Rec	NNC	1.677273e+04
1020	1993	4	Rec	NNC	3.725324e+03
1021	1994	1	Rec	NNC	0.000000e+00
1022	1994	2	Rec	NNC	1.311286e+05
1023	1994	3	Rec	NNC	1.101410e+05
1024	1994	4	Rec	NNC	2.273649e+04
1025	1995	1	Rec	NNC	9.011241e+02
1026	1995	2	Rec	NNC	1.396278e+05
1027	1995	3	Rec	NNC	1.128793e+05
1028	1995	4	Rec	NNC	1.893502e+04
1029	1996	1	Rec	NNC	0.000000e+00
1030	1996	2	Rec	NNC	1.076565e+05
1031	1996	3	Rec	NNC	6.420212e+04
1032	1996	4	Rec	NNC	2.472091e+04
1033	1997	1	Rec	NNC	0.000000e+00
1034	1997	2	Rec	NNC	4.358563e+05
1035	1997	3	Rec	NNC	1.733239e+05
1036	1997	4	Rec	NNC	5.107759e+04
1037	1998	1	Rec	NNC	0.000000e+00
1038	1998	2	Rec	NNC	2.813807e+05
1039	1998	3	Rec	NNC	8.236200e+04
1040	1998	4	Rec	NNC	4.856163e+04
1041	1999	1	Rec	NNC	0.000000e+00
1042	1999	2	Rec	NNC	7.658788e+05

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1043 1999   3     Rec  NNC                      1.105604e+05
1044 1999   4     Rec  NNC                      4.383661e+05
1045 2000   1     Rec  NNC                      0.000000e+00
1046 2000   2     Rec  NNC                      4.822238e+05
1047 2000   3     Rec  NNC                      3.086571e+05
1048 2000   4     Rec  NNC                      3.609943e+04
1049 2001   1     Rec  NNC                      0.000000e+00
1050 2001   2     Rec  NNC                      1.306376e+06
1051 2001   3     Rec  NNC                      4.407184e+05
1052 2001   4     Rec  NNC                      1.362622e+04
1053 2002   1     Rec  NNC                      0.000000e+00
1054 2002   2     Rec  NNC                      5.714948e+05
1055 2002   3     Rec  NNC                      2.149395e+05
1056 2002   4     Rec  NNC                      8.549641e+04
1057 2003   1     Rec  NNC                      0.000000e+00
1058 2003   2     Rec  NNC                      6.097702e+05
1059 2003   3     Rec  NNC                      2.393101e+05
1060 2003   4     Rec  NNC                      1.911989e+04
1061 2004   1     Rec  NNC                      0.000000e+00
1062 2004   2     Rec  NNC                      1.130043e+05
1063 2004   3     Rec  NNC                      4.759809e+05
1064 2004   4     Rec  NNC                      2.136272e+03
1065 2005   1     Rec  NNC                      2.136272e+03
1066 2005   2     Rec  NNC                      5.906869e+05
1067 2005   3     Rec  NNC                      2.595178e+05
1068 2005   4     Rec  NNC                      5.415554e+04
1069 2006   1     Rec  NNC                      0.000000e+00
1070 2006   2     Rec  NNC                      6.079730e+05
1071 2006   3     Rec  NNC                      1.069085e+05
1072 2006   4     Rec  NNC                      2.871739e+04
1073 2007   1     Rec  NNC                      5.942542e+03
1074 2007   2     Rec  NNC                      7.446139e+05
1075 2007   3     Rec  NNC                      4.911765e+05
1076 2007   4     Rec  NNC                      1.638259e+05
1077 2008   1     Rec  NNC                      8.401426e+04
1078 2008   2     Rec  NNC                      1.520610e+05
1079 2008   3     Rec  NNC                      7.150417e+04
1080 2008   4     Rec  NNC                      0.000000e+00
1081 2009   1     Rec  NNC                      0.000000e+00
1082 2009   2     Rec  NNC                      1.264628e+06
1083 2009   3     Rec  NNC                      1.087706e+06
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1084 2009   4      Rec   NNC                                  6.844584e+03
1085 2010   1      Rec   NNC                                  2.232589e+04
1086 2010   2      Rec   NNC                                  6.534632e+05
1087 2010   3      Rec   NNC                                  4.043579e+05
1088 2010   4      Rec   NNC                                  3.033275e+04
1089 2011   1      Rec   NNC                                  0.000000e+00
1090 2011   2      Rec   NNC                                  1.069911e+06
1091 2011   3      Rec   NNC                                  4.698892e+05
1092 2011   4      Rec   NNC                                  5.235156e+05
1093 2012   1      Rec   NNC                                  2.302064e+03
1094 2012   2      Rec   NNC                                  4.907943e+05
1095 2012   3      Rec   NNC                                  3.902222e+05
1096 2012   4      Rec   NNC                                  1.761655e+05
1097 2013   1      Rec   NNC                                  2.312303e+04
1098 2013   2      Rec   NNC                                  4.094121e+05
1099 2013   3      Rec   NNC                                  1.233905e+05
1100 2013   4      Rec   NNC                                  7.874372e+03
1101 2014   1      Rec   NNC                                  0.000000e+00
1102 2014   2      Rec   NNC                                  1.110407e+06
1103 2014   3      Rec   NNC                                  1.588618e+05
1104 2014   4      Rec   NNC                                  2.043554e+04
1105 2015   1      Rec   NNC                                  0.000000e+00
1106 2015   2      Rec   NNC                                  2.063566e+06
1107 2015   3      Rec   NNC                                  1.061391e+06
1108 2015   4      Rec   NNC                                  3.802375e+05
1109 2016   1      Rec   NNC                                  0.000000e+00
1110 2016   2      Rec   NNC                                  1.047769e+06
1111 2016   3      Rec   NNC                                  6.005003e+05
1112 2016   4      Rec   NNC                                  5.639231e+04
1113 2017   1      Rec   NNC                                  0.000000e+00
1114 2017   2      Rec   NNC                                  4.679636e+05
1115 2017   3      Rec   NNC                                  3.251755e+05
1116 2017   4      Rec   NNC                                  8.185031e+03
1117 2018   1      Rec   NNC                                  0.000000e+00
1118 2018   2      Rec   NNC                                  6.791719e+05
1119 2018   3      Rec   NNC                                  3.564936e+05
1120 2018   4      Rec   NNC                                  3.120586e+05
1121 2019   1      Rec   NNC                                  0.000000e+00
1122 2019   2      Rec   NNC                                  1.086972e+06
1123 2019   3      Rec   NNC                                  3.921515e+05
1124 2019   4      Rec   NNC                                  2.081160e+05
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1125 2020   1     Rec  NNC                       0.000000e+00
1126 2020   2     Rec  NNC                       6.442826e+05
1127 2020   3     Rec  NNC                       4.355690e+05
1128 2020   4     Rec  NNC                       9.393537e+03
1129 2021   1     Rec  NNC                       0.000000e+00
1130 2021   2     Rec  NNC                       3.354591e+05
1131 2021   3     Rec  NNC                       2.612134e+05
1132 2021   4     Rec  NNC                       0.000000e+00
1133 2022   1     Rec  NNC                       0.000000e+00
1134 2022   2     Rec  NNC                       1.095702e+05
1135 2022   3     Rec  NNC                       3.506203e+04
1136 2022   4     Rec  NNC                       6.676231e+04
1137 2023   1     Rec  NNC                       0.000000e+00
1138 2023   2     Rec  NNC                       1.497840e+05
1139 2023   3     Rec  NNC                       2.540520e+04
1140 2023   4     Rec  NNC                       0.000000e+00
1141 2024   1     Rec  NNC                       0.000000e+00
1142 2024   2     Rec  NNC                       8.349211e+04
1143 2024   3     Rec  NNC                       2.281517e+04
1144 2024   4     Rec  NNC                       0.000000e+00
1145 2025   1     Rec  NNC                       0.000000e+00
1146 2025   2     Rec  NNC                                 NA
1147 2025   3     Rec  NNC                                 NA
1148 2025   4     Rec  NNC                                 NA
1149 1985   1     Rec  VBM                       0.000000e+00
1150 1985   2     Rec  VBM                       1.129443e+05
1151 1985   3     Rec  VBM                       0.000000e+00
1152 1985   4     Rec  VBM                       0.000000e+00
1153 1986   1     Rec  VBM                       0.000000e+00
1154 1986   2     Rec  VBM                       7.339022e+04
1155 1986   3     Rec  VBM                       0.000000e+00
1156 1986   4     Rec  VBM                       3.055492e+04
1157 1987   1     Rec  VBM                       0.000000e+00
1158 1987   2     Rec  VBM                       4.206392e+04
1159 1987   3     Rec  VBM                       0.000000e+00
1160 1987   4     Rec  VBM                       1.007585e+04
1161 1988   1     Rec  VBM                       0.000000e+00
1162 1988   2     Rec  VBM                       2.932505e+04
1163 1988   3     Rec  VBM                       0.000000e+00
1164 1988   4     Rec  VBM                       4.241116e+03
1165 1989   1     Rec  VBM                       0.000000e+00
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1166 1989   2     Rec  VBM                      6.821138e+05
1167 1989   3     Rec  VBM                      6.273420e+03
1168 1989   4     Rec  VBM                      3.085581e+05
1169 1990   1     Rec  VBM                      0.000000e+00
1170 1990   2     Rec  VBM                      1.469085e+05
1171 1990   3     Rec  VBM                      0.000000e+00
1172 1990   4     Rec  VBM                      3.625165e+05
1173 1991   1     Rec  VBM                      0.000000e+00
1174 1991   2     Rec  VBM                      3.136398e+05
1175 1991   3     Rec  VBM                      0.000000e+00
1176 1991   4     Rec  VBM                      1.511470e+05
1177 1992   1     Rec  VBM                      0.000000e+00
1178 1992   2     Rec  VBM                      8.693982e+04
1179 1992   3     Rec  VBM                      1.026340e+04
1180 1992   4     Rec  VBM                      4.120940e+05
1181 1993   1     Rec  VBM                      0.000000e+00
1182 1993   2     Rec  VBM                      6.988338e+05
1183 1993   3     Rec  VBM                      3.495753e+03
1184 1993   4     Rec  VBM                      1.696190e+05
1185 1994   1     Rec  VBM                      0.000000e+00
1186 1994   2     Rec  VBM                      5.119812e+05
1187 1994   3     Rec  VBM                      2.273618e+03
1188 1994   4     Rec  VBM                      8.021262e+04
1189 1995   1     Rec  VBM                      0.000000e+00
1190 1995   2     Rec  VBM                      5.290627e+05
1191 1995   3     Rec  VBM                      8.023987e+04
1192 1995   4     Rec  VBM                      0.000000e+00
1193 1996   1     Rec  VBM                      0.000000e+00
1194 1996   2     Rec  VBM                      4.084946e+05
1195 1996   3     Rec  VBM                      0.000000e+00
1196 1996   4     Rec  VBM                      7.761164e+04
1197 1997   1     Rec  VBM                      0.000000e+00
1198 1997   2     Rec  VBM                      1.498605e+05
1199 1997   3     Rec  VBM                      0.000000e+00
1200 1997   4     Rec  VBM                      4.801143e+04
1201 1998   1     Rec  VBM                      0.000000e+00
1202 1998   2     Rec  VBM                      4.708166e+05
1203 1998   3     Rec  VBM                      0.000000e+00
1204 1998   4     Rec  VBM                      1.359527e+05
1205 1999   1     Rec  VBM                      0.000000e+00
1206 1999   2     Rec  VBM                      2.411140e+05
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1207 1999   3     Rec  VBM                                2.572690e+04
1208 1999   4     Rec  VBM                                2.567828e+05
1209 2000   1     Rec  VBM                                0.000000e+00
1210 2000   2     Rec  VBM                                1.440373e+06
1211 2000   3     Rec  VBM                                0.000000e+00
1212 2000   4     Rec  VBM                                1.681063e+05
1213 2001   1     Rec  VBM                                0.000000e+00
1214 2001   2     Rec  VBM                                2.879419e+05
1215 2001   3     Rec  VBM                                5.977503e+04
1216 2001   4     Rec  VBM                                2.490633e+04
1217 2002   1     Rec  VBM                                0.000000e+00
1218 2002   2     Rec  VBM                                6.060759e+05
1219 2002   3     Rec  VBM                                1.599474e+05
1220 2002   4     Rec  VBM                                9.663327e+05
1221 2003   1     Rec  VBM                                0.000000e+00
1222 2003   2     Rec  VBM                                1.223911e+05
1223 2003   3     Rec  VBM                                0.000000e+00
1224 2003   4     Rec  VBM                                7.823895e+05
1225 2004   1     Rec  VBM                                0.000000e+00
1226 2004   2     Rec  VBM                                1.283020e+05
1227 2004   3     Rec  VBM                                0.000000e+00
1228 2004   4     Rec  VBM                                5.823660e+04
1229 2005   1     Rec  VBM                                0.000000e+00
1230 2005   2     Rec  VBM                                2.756822e+05
1231 2005   3     Rec  VBM                                0.000000e+00
1232 2005   4     Rec  VBM                                2.135543e+05
1233 2006   1     Rec  VBM                                0.000000e+00
1234 2006   2     Rec  VBM                                6.831538e+05
1235 2006   3     Rec  VBM                                1.111672e+05
1236 2006   4     Rec  VBM                                4.201761e+05
1237 2007   1     Rec  VBM                                0.000000e+00
1238 2007   2     Rec  VBM                                2.089240e+05
1239 2007   3     Rec  VBM                                0.000000e+00
1240 2007   4     Rec  VBM                                2.784509e+04
1241 2008   1     Rec  VBM                                0.000000e+00
1242 2008   2     Rec  VBM                                6.659730e+04
1243 2008   3     Rec  VBM                                2.248205e+04
1244 2008   4     Rec  VBM                                7.739115e+03
1245 2009   1     Rec  VBM                                0.000000e+00
1246 2009   2     Rec  VBM                                1.573958e+06
1247 2009   3     Rec  VBM                                3.543890e+03
```

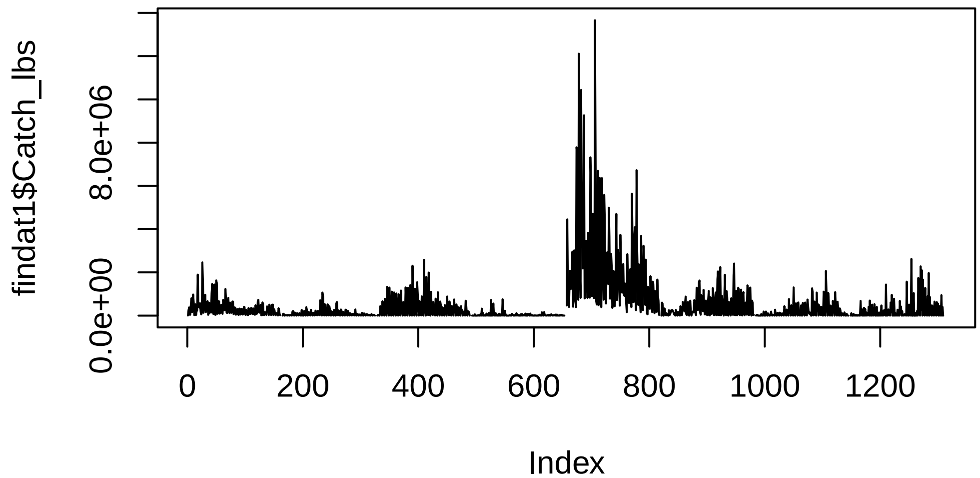
1248	2009	4	Rec	VBM	3.041695e+04
1249	2010	1	Rec	VBM	0.000000e+00
1250	2010	2	Rec	VBM	5.114340e+05
1251	2010	3	Rec	VBM	5.630438e+03
1252	2010	4	Rec	VBM	2.003465e+05
1253	2011	1	Rec	VBM	0.000000e+00
1254	2011	2	Rec	VBM	2.626410e+06
1255	2011	3	Rec	VBM	0.000000e+00
1256	2011	4	Rec	VBM	0.000000e+00
1257	2012	1	Rec	VBM	0.000000e+00
1258	2012	2	Rec	VBM	1.041694e+06
1259	2012	3	Rec	VBM	0.000000e+00
1260	2012	4	Rec	VBM	0.000000e+00
1261	2013	1	Rec	VBM	0.000000e+00
1262	2013	2	Rec	VBM	1.374911e+05
1263	2013	3	Rec	VBM	0.000000e+00
1264	2013	4	Rec	VBM	2.270012e+05
1265	2014	1	Rec	VBM	0.000000e+00
1266	2014	2	Rec	VBM	1.746051e+06
1267	2014	3	Rec	VBM	1.453196e+05
1268	2014	4	Rec	VBM	1.489590e+05
1269	2015	1	Rec	VBM	0.000000e+00
1270	2015	2	Rec	VBM	2.279425e+06
1271	2015	3	Rec	VBM	2.682508e+05
1272	2015	4	Rec	VBM	2.098120e+06
1273	2016	1	Rec	VBM	0.000000e+00
1274	2016	2	Rec	VBM	1.667758e+06
1275	2016	3	Rec	VBM	2.730686e+05
1276	2016	4	Rec	VBM	2.453000e+05
1277	2017	1	Rec	VBM	0.000000e+00
1278	2017	2	Rec	VBM	1.287459e+06
1279	2017	3	Rec	VBM	2.537905e+04
1280	2017	4	Rec	VBM	5.139850e+05
1281	2018	1	Rec	VBM	0.000000e+00
1282	2018	2	Rec	VBM	8.945892e+05
1283	2018	3	Rec	VBM	5.947017e+03
1284	2018	4	Rec	VBM	1.970588e+06
1285	2019	1	Rec	VBM	0.000000e+00
1286	2019	2	Rec	VBM	8.636894e+05
1287	2019	3	Rec	VBM	1.059053e+05
1288	2019	4	Rec	VBM	4.708263e+05

```
1289 2020   1      Rec   VBM                     0.000000e+00
1290 2020   2      Rec   VBM                     4.699253e+05
1291 2020   3      Rec   VBM                     7.429021e+03
1292 2020   4      Rec   VBM                     2.117452e+05
1293 2021   1      Rec   VBM                     0.000000e+00
1294 2021   2      Rec   VBM                     6.502247e+05
1295 2021   3      Rec   VBM                     1.280529e+04
1296 2021   4      Rec   VBM                     4.192489e+05
1297 2022   1      Rec   VBM                     9.898678e+02
1298 2022   2      Rec   VBM                     6.221143e+05
1299 2022   3      Rec   VBM                     2.531032e+04
1300 2022   4      Rec   VBM                     5.588537e+05
1301 2023   1      Rec   VBM                     0.000000e+00
1302 2023   2      Rec   VBM                     4.341456e+05
1303 2023   3      Rec   VBM                     3.106470e+03
1304 2023   4      Rec   VBM                     4.352388e+05
1305 2024   1      Rec   VBM                     0.000000e+00
1306 2024   2      Rec   VBM                     9.457560e+05
1307 2024   3      Rec   VBM                     7.382052e+02
1308 2024   4      Rec   VBM                     4.001127e+05
1309 2025   1      Rec   VBM                     0.000000e+00
1310 2025   2      Rec   VBM                               NA
1311 2025   3      Rec   VBM                               NA
1312 2025   4      Rec   VBM                               NA
```

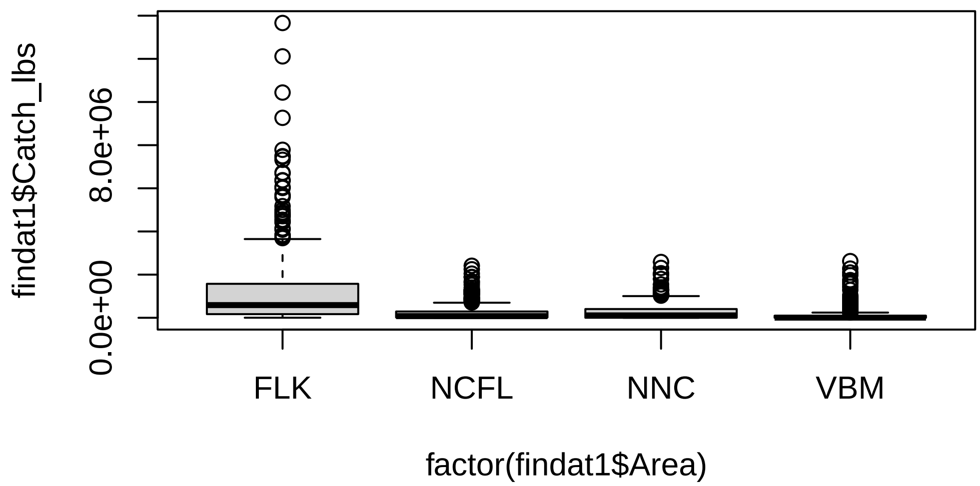
```r
names(findat1) <- c("Year", "Quarter", "Fleet", "Area", "Catch_lbs")

head(findat1)
```

```
  Year Quarter   Fleet Area Catch_lbs
1 1985       1 ForHire  FLK   4462.09
2 1985       2 ForHire  FLK 253487.38
3 1985       3 ForHire  FLK 380811.37
4 1985       4 ForHire  FLK  41469.99
5 1986       1 ForHire  FLK  40545.41
6 1986       2 ForHire  FLK 501184.61
```

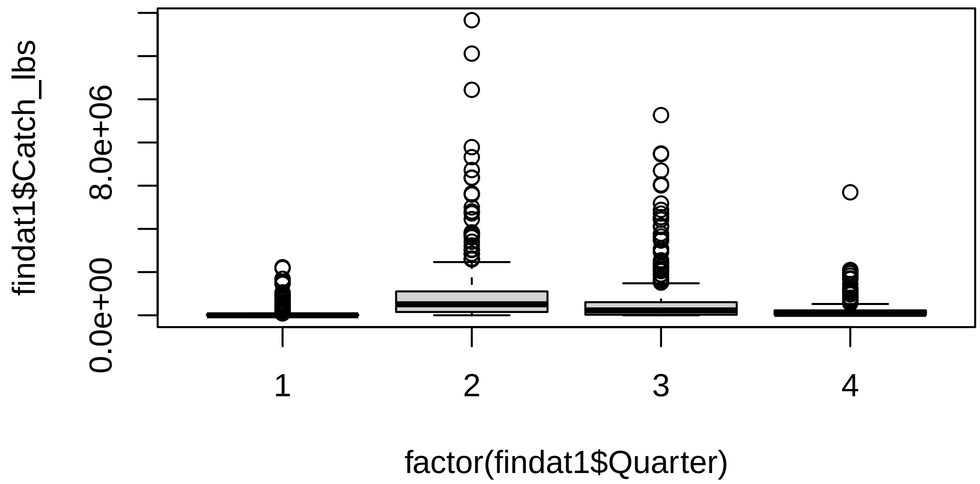
```r
plot(findat1$Catch_lbs, type = "l")
```



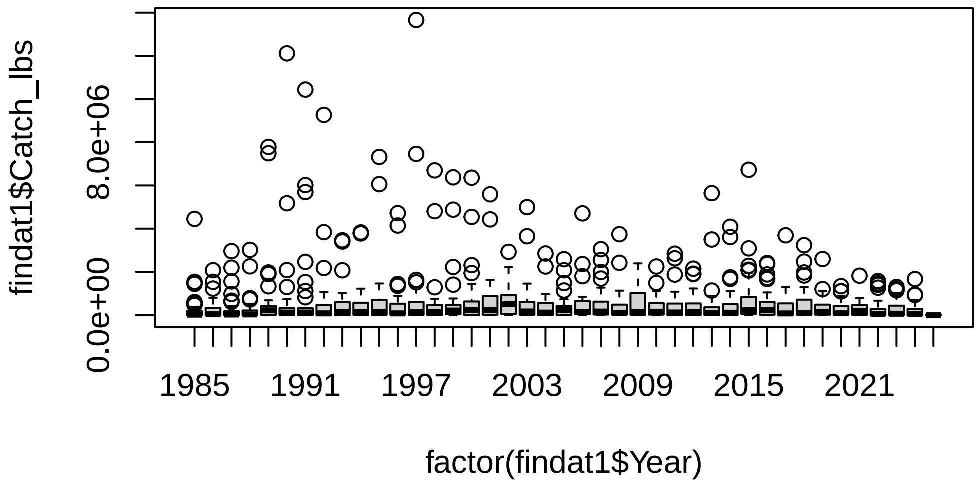

```r
plot(findat1$Catch_lbs ~ factor(findat1$Area))
```



```
plot(findat1$Catch_lbs ~ factor(findat1$Quarter))
```




```
plot(findat1$Catch_lbs ~ factor(findat1$Year))
```




```
findat1 <- findat1[which(findat1$Year <= 2022), ]
findat1 <- findat1[which(findat1$Year >= 1986), ]

apply(findat1, 2, table)
```

```
$Year

1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001
  32   32   32   32   32   32   32   32   32   32   32   32   32   32   32   32
2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017
  32   32   32   32   32   32   32   32   32   32   32   32   32   32   32   32
2018 2019 2020 2021 2022
  32   32   32   32   32

$Quarter

  1   2   3   4
296 296 296 296

$Fleet
```

```
ForHire     Rec
    592     592

$Area

 FLK NCFL  NNC  VBM
 296  296  296  296

$Catch_lbs

0.000000e+00 1.004109e+05 1.004864e+06 1.006040e+06 1.007585e+04 1.007698e+04
         114            1            1            1            1            1
1.011253e+05 1.014313e+02 1.015929e+05 1.017529e+05 1.018500e+04 1.026340e+04
           1            1            1            1            1            1
1.026905e+06 1.027396e+06 1.036612e+04 1.036735e+04 1.039095e+06 1.041694e+06
           1            1            1            1            1            1
1.043543e+05 1.043784e+07 1.047769e+06 1.054628e+04 1.057200e+05 1.058325e+05
           1            1            1            1            1            1
1.059053e+05 1.059973e+04 1.060551e+06 1.061391e+06 1.063765e+05 1.064174e+03
           1            1            1            1            1            1
1.067933e+06 1.069085e+05 1.069911e+06 1.070780e+06 1.070805e+05 1.074642e+06
           1            1            1            1            1            1
1.075557e+00 1.076565e+05 1.077096e+05 1.077716e+05 1.078562e+03 1.081341e+06
           1            1            1            1            1            1
1.084238e+03 1.086972e+06 1.087706e+06 1.088889e+06 1.092633e+05 1.093047e+06
           1            1            1            1            1            1
1.095702e+05 1.098993e+01 1.101410e+05 1.103121e+06 1.105212e+06 1.105604e+05
           1            1            1            1            1            1
1.110407e+06 1.111538e+06 1.111672e+05 1.113341e+06 1.114007e+02 1.118934e+05
           1            1            1            1            1            1
1.119818e+06 1.121239e+06 1.125717e+05 1.126962e+05 1.128793e+05 1.130043e+05
           1            1            1            1            1            1
1.130458e+06 1.133952e+06 1.135865e+05 1.137113e+06 1.138712e+03 1.139008e+04
           1            1            1            1            1            1
1.143834e+04 1.145773e+06 1.147492e+05 1.151956e+05 1.152953e+05 1.154803e+05
           1            1            1            1            1            1
1.159216e+02 1.159326e+05 1.163134e+06 1.164199e+04 1.170229e+04 1.172606e+04
           1            1            1            1            1            1
1.173219e+05 1.178413e+04 1.180400e+03 1.183999e+03 1.185557e+04 1.196275e+06
           1            1            1            1            1            1
```

```
1.197307e+05 1.200051e+05 1.201582e+05 1.202456e+06 1.203616e+05 1.211452e+07 
           1            1            1            1            1            1 
1.214296e+06 1.223023e+06 1.223197e+06 1.223911e+05 1.225359e+06 1.229422e+06 
           1            1            1            1            1            1 
1.233226e+06 1.233905e+05 1.240299e+05 1.240658e+04 1.241709e+04 1.246703e+06 
           1            1            1            1            1            1 
1.250031e+05 1.250153e+03 1.255034e+02 1.264628e+06 1.267020e+06 1.268359e+05 
           1            1            1            1            1            1 
1.269099e+06 1.270259e+03 1.280529e+04 1.283020e+05 1.286262e+06 1.287459e+06 
           1            1            1            1            1            1 
1.292812e+06 1.292909e+06 1.298744e+06 1.305418e+06 1.305756e+00 1.306186e+05 
           1            1            1            1            1            1 
1.306376e+06 1.311151e+05 1.311286e+05 1.327154e+05 1.329583e+05 1.331147e+06 
           1            1            1            1            1            1 
1.336144e+06 1.339044e+01 1.344365e+05 1.351256e+05 1.352282e+06 1.353495e+05 
           1            1            1            1            1            1 
1.353952e+04 1.356446e+04 1.359527e+05 1.361174e+05 1.362622e+04 1.366213e+07 
           1            1            1            1            1            1 
1.367443e+05 1.374911e+05 1.379817e+04 1.385021e+04 1.387479e+05 1.390055e+05 
           1            1            1            1            1            1 
1.393842e+05 1.396278e+05 1.398654e+06 1.399500e+05 1.400650e+05 1.406221e+05 
           1            1            1            1            1            1 
1.408221e+06 1.413382e+04 1.425417e+03 1.433616e-01 1.439914e+06 1.440373e+06 
           1            1            1            1            1            1 
1.443916e+04 1.444289e+04 1.447126e+04 1.453196e+05 1.455036e+00 1.456263e+06 
           1            1            1            1            1            1 
1.458389e+05 1.459837e+05 1.461029e+06 1.465598e+05 1.465611e+03 1.469085e+05 
           1            1            1            1            1            1 
1.470855e+06 1.476583e+05 1.478184e+01 1.479721e+04 1.480608e+04 1.481262e+06 
           1            1            1            1            1            1 
1.487555e+06 1.489590e+05 1.489910e+05 1.493810e+05 1.493999e+05 1.494802e+05 
           1            1            1            1            1            1 
1.498605e+05 1.499484e+05 1.501280e+05 1.502980e+04 1.505553e+04 1.505825e+04 
           1            1            1            1            1            1 
1.511470e+05 1.511596e+05 1.512397e+05 1.512896e+04 1.514418e+06 1.516255e+04 
           1            1            1            1            1            1 
1.519864e+05 1.520610e+05 1.526796e+06 1.530429e+06 1.535951e+04 1.537010e+06 
           1            1            1            1            1            1 
1.537656e+05 1.538532e+04 1.539632e+04 1.544282e+05 1.545059e+06 1.550273e+05 
           1            1            1            1            1            1 
1.553554e+04 1.556368e+04 1.556820e+06 1.566620e+04 1.566697e+05 1.573958e+06 
```

```
           1            1            1            1            1            1 
1.574686e+05 1.574723e+06 1.575236e+05 1.582543e+05 1.585451e+05 1.585921e+03 
           1            1            1            1            1            1 
1.588618e+05 1.588620e+03 1.590883e+03 1.599474e+05 1.599922e+04 1.617787e+04 
           1            1            1            1            1            1 
1.626879e+06 1.629481e+06 1.633176e+05 1.634984e+05 1.638259e+05 1.644082e+04 
           1            1            1            1            1            1 
1.653072e+05 1.659762e+05 1.665554e+05 1.667679e+03 1.667758e+06 1.672854e+05 
           1            1            1            1            1            1 
1.674867e+06 1.677273e+04 1.681063e+05 1.686078e+04 1.687930e+06 1.689311e+06 
           1            1            1            1            1            1 
1.689554e+05 1.694061e+05 1.696190e+05 1.707999e+06 1.718750e+06 1.719227e+05 
           1            1            1            1            1            1 
1.724731e+04 1.727708e+05 1.731352e+05 1.733239e+05 1.742143e+05 1.743354e+04 
           1            1            1            1            1            1 
1.744758e+05 1.746051e+06 1.747245e+05 1.756989e+03 1.761655e+05 1.778849e+05 
           1            1            1            1            1            1 
1.779384e+03 1.780131e+05 1.782641e+04 1.790000e+03 1.798123e+05 1.798733e+06 
           1            1            1            1            1            1 
1.805194e+04 1.819174e+05 1.821803e+06 1.827329e-01 1.831259e+06 1.834842e+04 
           1            1            1            1            1            1 
1.839464e+04 1.839589e+03 1.849069e+04 1.855573e+05 1.863238e+05 1.873414e+05 
           1            1            1            1            1            1 
1.874902e+06 1.877862e+05 1.880964e+06 1.882011e+05 1.885881e+04 1.893502e+04 
           1            1            1            1            1            1 
1.894418e+06 1.898385e+06 1.902211e-01 1.902856e+05 1.903206e+05 1.904235e+05 
           1            1            1            1            1            1 
1.909134e+05 1.911989e+04 1.912235e+05 1.913413e+06 1.918508e+05 1.942786e+03 
           1            1            1            1            1            1 
1.945322e+06 1.946561e-01 1.952106e+05 1.960574e+05 1.961057e+06 1.970588e+06 
           1            1            1            1            1            1 
1.972244e+06 1.988053e+03 1.993879e+06 1.996363e+05 1.996467e+05 2.001049e+05 
           1            1            1            1            1            1 
2.003465e+05 2.005246e+03 2.005999e+05 2.016812e+05 2.027951e+05 2.040101e+03 
           1            1            1            1            1            1 
2.043554e+04 2.046079e+06 2.048967e+05 2.063566e+06 2.068819e+04 2.072151e+04 
           1            1            1            1            1            1 
2.073350e+06 2.074780e+06 2.075606e+06 2.077107e+05 2.081160e+05 2.081267e+06 
           1            1            1            1            1            1 
2.086118e+05 2.089240e+05 2.098120e+06 2.109711e+05 2.113411e+05 2.117452e+05 
           1            1            1            1            1            1 
```

```
2.128542e+04 2.130953e+05 2.132708e+04 2.135543e+05 2.136272e+03 2.140903e+04
           1            1            1            1            2            1
2.144051e+04 2.144944e+05 2.147562e+06 2.149395e+05 2.178847e+06 2.191794e+04
           1            1            1            1            1            1
2.192920e+06 2.193667e+05 2.194358e+05 2.209559e+05 2.214085e+06 2.214669e+03
           1            1            1            1            1            1
2.224750e+06 2.232589e+04 2.244790e+05 2.245187e+04 2.247155e+06 2.248205e+04
           1            1            1            1            1            1
2.248755e+06 2.253282e+06 2.253837e+05 2.263707e+02 2.266328e+05 2.270012e+05
           1            1            1            1            1            1
2.273618e+03 2.273649e+04 2.276001e+04 2.279425e+06 2.281061e+01 2.290328e+04
           1            1            1            1            1            1
2.294885e+05 2.296609e+03 2.297842e+05 2.302064e+03 2.302774e+05 2.307627e+06
           1            1            1            1            1            1
2.310664e+06 2.312303e+04 2.312582e+05 2.315459e+05 2.325079e+05 2.329070e+04
           1            1            1            1            1            1
2.333242e+05 2.342309e+04 2.348243e+04 2.360906e+06 2.361142e+02 2.370240e+06
           1            1            1            1            1            1
2.370312e+04 2.391186e+06 2.393101e+05 2.400880e+02 2.401520e+04 2.411140e+05
           1            1            1            1            1            1
2.411574e+06 2.414829e+05 2.417873e+05 2.419131e+06 2.420522e+04 2.420607e+04
           1            1            1            1            1            1
2.427385e+04 2.433248e+05 2.436482e+04 2.437370e+05 2.440593e+03 2.442929e+05
           1            1            1            1            1            1
2.444075e+05 2.450390e-01 2.453000e+05 2.454607e+04 2.459965e+05 2.460745e+05
           1            1            1            1            1            1
2.463709e+06 2.469372e+02 2.472091e+04 2.474158e+02 2.474352e+06 2.490633e+04
           1            1            1            1            1            1
2.491889e+04 2.499202e+04 2.502125e+05 2.502685e+05 2.513409e+05 2.524921e+05
           1            1            1            1            1            1
2.531032e+04 2.533863e+05 2.533937e+04 2.537905e+04 2.545278e+06 2.547820e+04
           1            1            1            1            1            1
2.549586e+03 2.565000e+05 2.566053e+04 2.567828e+05 2.572690e+04 2.583884e+03
           1            1            1            1            1            1
2.585572e+06 2.590388e+04 2.593726e+06 2.595178e+05 2.595592e+05 2.602455e+02
           1            1            1            1            1            1
2.612134e+05 2.617131e+05 2.626410e+06 2.627216e+05 2.637966e+04 2.639086e+04
           1            1            1            1            1            1
2.655088e+04 2.665228e+03 2.668275e+05 2.669028e+04 2.674703e+03 2.682508e+05
           1            1            1            1            1            1
2.687381e+03 2.687809e+05 2.689442e+05 2.701667e+05 2.719131e+03 2.719797e+05
```

```
             1              1              1              1              1              1 
2.720876e+05 2.726127e+05 2.730686e+05 2.737798e+04 2.752532e+05 2.756822e+05 
             1              1              1              1              1              1 
2.759011e+05 2.762370e+05 2.766761e+05 2.770928e+03 2.784509e+04 2.795122e+04 
             1              1              1              1              1              1 
2.796967e+04 2.797176e+04 2.800349e+04 2.804606e+05 2.813807e+05 2.832672e+04 
             1              1              1              1              1              1 
2.835385e+04 2.838386e+05 2.840895e+06 2.843143e+05 2.849332e+06 2.856638e+05 
             1              1              1              1              1              1 
2.865057e+04 2.871739e+04 2.872233e+05 2.879419e+05 2.880463e+04 2.923724e+05 
             1              1              1              1              1              1 
2.925741e+06 2.932505e+04 2.940661e+04 2.945357e+05 2.947750e+05 2.961105e+03 
             1              1              1              1              1              1 
2.961240e+06 2.963646e+05 2.967570e+02 2.970883e+04 2.973395e+05 2.973731e+04 
             1              1              1              1              1              1 
2.982117e+03 2.983204e+03 2.984649e+05 2.988903e+03 2.993657e+05 3.009903e+05 
             1              1              1              1              1              1 
3.017922e+05 3.018472e+06 3.033275e+04 3.041695e+04 3.042971e+04 3.043074e+05 
             1              1              1              1              1              1 
3.047614e+05 3.048185e+06 3.049959e+02 3.055492e+04 3.063555e+05 3.073598e+04 
             1              1              1              1              1              1 
3.085426e+04 3.085581e+05 3.086404e+06 3.086571e+05 3.086638e+05 3.096802e+05 
             1              1              1              1              1              1 
3.120586e+05 3.136398e+05 3.145215e+05 3.157635e+05 3.176531e+03 3.213315e+05 
             1              1              1              1              1              1 
3.234459e+06 3.237039e+05 3.239231e+03 3.251755e+05 3.255974e+02 3.259995e+05 
             1              1              1              1              1              1 
3.274145e+05 3.281162e+05 3.287088e+05 3.291462e+05 3.321194e+04 3.323576e+05 
             1              1              1              1              1              1 
3.338597e+05 3.341788e+04 3.346052e+05 3.354591e+05 3.360800e+02 3.369815e+05 
             1              1              1              1              1              1 
3.386228e+05 3.394897e+02 3.404812e+06 3.407217e+05 3.416471e+03 3.419584e+05 
             1              1              1              1              1              1 
3.421213e+05 3.433560e+04 3.435103e+03 3.438865e+04 3.452582e+03 3.455564e+04 
             1              1              1              1              1              1 
3.461382e+06 3.474931e+02 3.488479e+04 3.493479e+04 3.495753e+03 3.496798e+06 
             1              1              1              1              1              1 
3.499724e+05 3.506203e+04 3.516103e+05 3.523793e+04 3.528256e+04 3.543890e+03 
             1              1              1              1              1              1 
3.551940e+04 3.564936e+05 3.577490e+02 3.588338e+04 3.594554e+04 3.594953e+04 
             1              1              1              1              1              1 
```

```
3.596215e+05 3.599090e+05 3.609216e+06 3.609943e+04 3.612587e+04 3.615959e+04
           1            1            1            1            1            1
3.625165e+05 3.634808e+05 3.634945e+04 3.636352e+03 3.644628e+02 3.649030e+06
           1            1            1            1            1            1
3.657414e+03 3.669715e+04 3.673921e+05 3.674186e+04 3.691597e+06 3.691695e+04
           1            1            1            1            1            1
3.698497e+03 3.720073e+05 3.723780e+04 3.725324e+03 3.734484e+04 3.741868e+06
           1            1            1            1            1            1
3.745493e+05 3.779174e+06 3.796263e+02 3.802375e+05 3.823569e+05 3.825881e+06
           1            1            1            1            1            1
3.829178e+05 3.833937e+03 3.838259e+06 3.840700e+03 3.854991e+04 3.859802e+05
           1            1            1            1            1            1
3.864672e+04 3.866868e+00 3.872523e+03 3.874707e+04 3.883957e+05 3.885919e+04
           1            1            1            1            1            1
3.896585e+05 3.902222e+05 3.905219e+05 3.921515e+05 3.926185e+05 3.938811e+05
           1            1            1            1            1            1
3.945315e+05 3.958740e+02 3.961182e+05 3.962056e+05 3.984227e+03 3.987688e+04
           1            1            1            1            1            1
4.020906e+05 4.023345e+04 4.032198e+05 4.041334e+05 4.043579e+05 4.048370e+05
           1            1            1            1            1            1
4.050165e+00 4.054541e+04 4.056151e-01 4.056591e+05 4.065881e+02 4.078345e+05
           1            1            1            1            1            1
4.084946e+05 4.087638e+06 4.094121e+05 4.098999e+05 4.100432e+04 4.103876e+02
           1            1            1            1            1            1
4.120940e+05 4.121178e+05 4.132299e+01 4.144946e+06 4.147245e+04 4.157946e+04
           1            1            1            1            1            1
4.160198e+05 4.176974e+03 4.188082e+02 4.192489e+05 4.201761e+05 4.206392e+04
           1            1            1            1            1            1
4.231196e+03 4.241116e+03 4.246640e+05 4.269701e+05 4.271605e+05 4.272998e+04
           1            1            1            1            1            1
4.281886e+04 4.284965e+05 4.295189e+05 4.308598e+05 4.318811e+02 4.320121e+05
           1            1            1            1            1            1
4.331248e+05 4.332058e+03 4.336462e+04 4.343268e+04 4.355690e+05 4.355714e+05
           1            1            1            1            1            1
4.356505e+04 4.358563e+05 4.377578e+05 4.383661e+05 4.407184e+05 4.428842e+06
           1            1            1            1            1            1
4.465400e+05 4.478344e+03 4.489343e+05 4.497292e+05 4.499024e+04 4.515808e+04
           1            1            1            1            1            1
4.518215e+03 4.540662e+05 4.543957e+06 4.566275e+05 4.592291e+05 4.599396e+05
           1            1            1            1            1            1
4.607402e+04 4.617975e+05 4.679636e+05 4.698892e+05 4.699253e+05 4.707929e+06
```

```
           1            1            1            1            1            1 
4.708166e+05 4.708263e+05 4.715174e+04 4.717398e+06 4.729552e+02 4.741635e+04 
           1            1            1            1            1            1 
4.745844e+03 4.759809e+05 4.783451e+05 4.788760e+04 4.798984e+04 4.801143e+04 
           1            1            1            1            1            1 
4.808060e+05 4.808405e+06 4.814084e+05 4.822238e+05 4.825515e+03 4.838705e+03 
           1            1            1            1            1            1 
4.841136e+03 4.854292e+04 4.856163e+04 4.865816e+05 4.872517e+05 4.880669e+06 
           1            1            1            1            1            1 
4.898075e+05 4.907943e+05 4.911765e+05 4.919460e+05 4.926664e+04 4.930504e-01 
           1            1            1            1            1            1 
4.941575e+03 4.946504e+02 4.958988e+04 4.962654e+05 4.977416e+04 4.980747e+05 
           1            1            1            1            1            1 
4.995939e+06 5.011846e+05 5.011873e+05 5.033407e+05 5.062703e-01 5.063335e+05 
           1            1            1            1            1            1 
5.067091e+05 5.086524e+04 5.096188e+05 5.105764e+03 5.107759e+04 5.112349e+04 
           1            1            1            1            1            1 
5.113129e+05 5.114340e+05 5.119812e+05 5.126049e+05 5.132135e+05 5.139850e+05 
           1            1            1            1            1            1 
5.174288e+05 5.174374e+06 5.197939e+05 5.198576e+05 5.213514e+05 5.235156e+05 
           1            1            1            1            1            1 
5.236336e+05 5.237065e+05 5.238812e+03 5.279053e+05 5.279255e+05 5.290627e+05 
           1            1            1            1            1            1 
5.313994e+04 5.314265e+05 5.318615e+05 5.327003e+04 5.340980e+03 5.351815e+03 
           1            1            1            1            1            1 
5.359988e+03 5.384247e+05 5.388107e+02 5.397084e+04 5.412456e+05 5.415554e+04 
           1            1            1            1            1            1 
5.417106e+05 5.418005e+03 5.468464e+04 5.481602e+04 5.518140e+02 5.576250e+05 
           1            1            1            1            1            1 
5.587967e+04 5.588537e+05 5.589528e+06 5.604508e+02 5.608290e+05 5.630438e+03 
           1            1            1            1            1            1 
5.630532e+05 5.639231e+04 5.640033e+04 5.640292e+04 5.642923e+06 5.659894e+04 
           1            1            1            1            1            1 
5.660774e+05 5.694630e+06 5.695040e-01 5.714948e+05 5.723903e+03 5.730691e+05 
           1            1            1            1            1            1 
5.735168e+05 5.754550e+04 5.812551e+04 5.823660e+04 5.832363e+05 5.835857e+05 
           1            1            1            1            1            1 
5.843752e+05 5.880012e+04 5.903592e+05 5.906869e+05 5.918365e+05 5.934543e+05 
           1            1            1            1            1            1 
5.942542e+03 5.947017e+03 5.947019e+05 5.977503e+04 5.990593e+04 6.005003e+05 
           1            1            1            1            1            1 
```

```
6.015731e+06 6.019786e+03 6.037893e+05 6.059479e+06 6.060759e+05 6.070687e+05
           1            1            1            1            1            1
6.071702e+05 6.079730e+05 6.087522e+05 6.097702e+05 6.103580e+04 6.117980e+05
           1            1            1            1            1            1
6.137384e+04 6.161745e+03 6.204025e+03 6.215559e+05 6.221143e+05 6.222483e+01
           1            1            1            1            1            1
6.223060e+05 6.246935e+04 6.260843e+05 6.272307e+05 6.273420e+03 6.274736e+05
           1            1            1            1            1            1
6.296220e+05 6.313726e+04 6.328460e+05 6.344073e+05 6.355006e+05 6.358022e+06
           1            1            1            1            1            1
6.375714e+06 6.394769e+03 6.398102e+02 6.420212e+04 6.442826e+05 6.480516e+02
           1            1            1            1            1            1
6.486364e+04 6.495267e+02 6.502247e+05 6.507979e+00 6.529994e+05 6.534632e+05
           1            1            1            1            1            1
6.537249e+05 6.539262e+05 6.548437e+05 6.563503e+05 6.606708e+04 6.613227e+05
           1            1            1            1            1            1
6.613425e+04 6.614244e+04 6.634651e+05 6.657709e+05 6.659730e+04 6.676231e+04
           1            1            1            1            1            1
6.692056e+05 6.697606e+06 6.713363e+05 6.722192e+05 6.725790e+06 6.743990e+05
           1            1            1            1            1            1
6.761108e+02 6.791719e+05 6.803199e+05 6.804028e+02 6.818692e+03 6.821138e+05
           1            1            1            1            1            1
6.831538e+05 6.841681e+04 6.844584e+03 6.873502e+02 6.907945e+04 6.912068e+05
           1            1            1            1            1            1
6.914937e+04 6.945761e+03 6.980021e+04 6.988338e+05 6.991629e+04 6.994376e-01
           1            1            1            1            1            1
6.996628e+05 7.039843e+03 7.042666e+05 7.075256e+04 7.099938e+05 7.150417e+04
           1            1            1            1            1            1
7.160624e+04 7.170208e+05 7.202998e+05 7.204270e+05 7.207076e+05 7.239204e+02
           1            1            1            1            1            1
7.241465e+05 7.265192e+05 7.268633e+04 7.310031e+05 7.318250e+05 7.321132e+06
           1            1            1            1            1            1
7.334898e+05 7.339022e+04 7.347928e+05 7.388486e+04 7.402020e+04 7.412236e+04
           1            1            1            1            1            1
7.429021e+03 7.446139e+05 7.460292e+06 7.471727e+04 7.473885e+04 7.490678e+06
           1            1            1            1            1            1
7.493783e+03 7.496388e+05 7.553346e+05 7.596949e+05 7.634698e+02 7.648718e+04
           1            1            1            1            1            1
7.658788e+05 7.686783e+03 7.692057e+04 7.704207e+05 7.714475e+05 7.739115e+03
           1            1            1            1            1            1
7.761164e+04 7.773956e+04 7.785107e+06 7.786410e+05 7.807843e+05 7.823895e+05
```

```
           1            1            1            1            1            1 
7.824121e+02 7.827342e+05 7.848983e+05 7.853118e+05 7.868668e+04 7.870523e+05 
           1            1            1            1            1            1 
7.874372e+03 7.924758e+04 7.924777e+03 7.943520e+03 7.945853e+05 7.951533e+04 
           1            1            1            1            1            1 
7.965220e+01 7.976476e+03 8.021262e+04 8.023987e+04 8.024703e+05 8.049080e+04 
           1            1            1            1            1            1 
8.056149e+04 8.058665e+04 8.061179e+05 8.064427e+01 8.083241e+05 8.143585e+05 
           1            1            1            2            1            1 
8.152937e+04 8.185031e+03 8.204078e+05 8.213613e+03 8.213770e+04 8.214989e+05 
           1            1            1            1            1            1 
8.224854e+04 8.229003e+05 8.229340e+05 8.236200e+04 8.270966e+04 8.304530e+05 
           1            1            1            1            1            1 
8.315590e+04 8.316655e+04 8.379875e+04 8.401426e+04 8.427480e+02 8.428625e+04 
           1            1            1            1            1            1 
8.440390e+04 8.464547e+04 8.469482e+04 8.487273e+05 8.544770e+04 8.549641e+04 
           1            1            1            1            1            1 
8.570458e+04 8.596577e+05 8.603992e+02 8.617154e+04 8.636894e+05 8.680582e+04 
           1            1            1            1            1            1 
8.693982e+04 8.708714e+05 8.714024e+02 8.739791e+05 8.746241e+05 8.761344e+04 
           1            1            1            1            1            1 
8.768057e+04 8.796385e+05 8.825788e+05 8.873625e+05 8.876937e+05 8.883334e+04 
           1            1            1            1            1            1 
8.914606e+02 8.934973e+05 8.944467e+05 8.945892e+05 8.981145e+04 8.996649e+03 
           1            1            1            1            1            1 
9.008397e+04 9.011241e+02 9.044386e+03 9.085066e+04 9.086961e+04 9.104998e+00 
           1            1            1            1            1            1 
9.123982e+01 9.128843e+04 9.129205e+03 9.135979e+02 9.141962e+05 9.178266e+05 
           1            1            1            1            1            1 
9.207619e+03 9.266735e+06 9.367454e+05 9.376058e+02 9.387198e+03 9.393537e+03 
           1            1            1            1            1            1 
9.417821e+04 9.589750e+05 9.611454e+04 9.663327e+05 9.670219e+04 9.672219e+05 
           1            1            1            1            1            1 
9.679768e+04 9.688738e+05 9.702987e+04 9.711731e+04 9.717851e+05 9.739787e+05 
           1            1            1            1            1            1 
9.786403e+03 9.787720e+05 9.818841e+05 9.819018e+04 9.832395e+04 9.839474e+02 
           1            1            1            1            1            1 
9.855716e+04 9.876608e+00 9.896425e+05 9.898678e+02 9.909438e+04 9.912037e+01 
           1            1            1            1            1            1 
9.971934e+05 
           1 
```

```r
findat <- findat[which(findat$year >=1986 & findat$year <= 2022), ]
#check that the numbers are exactly the same
tapply(findat1$Catch_lbs, list(findat1$Area, findat1$Fleet), sum, na.rm = T)
```

```
      ForHire       Rec
FLK  54012811 357091651
NCFL 16532764  65643193
NNC  56097190  32197622
VBM   5736592  37749487
```

```r
colSums(findat[3:10], na.rm = T)
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM
 54012811  16532764  56097190   5736592 357091651  65643193  32197622  37749487
```

```r
write.csv(findat1, file = "data/rec_formattedTom.csv", row.names = FALSE)
```

3 Analysis of international landings estimates from Sea Around Us database

Here we will explore commercial and recreational landings data as estimated by the Sea Around Us Project. This data set includes reported data for fisheries worldwide as well as reconstructed estimates for unreported landings. For each country, project scientists assemble available data on landings and use additional sociological and fishing data plus expert information and opinion, to produce their best estimates of total landings from all fishing sectors and for all species globally.

3.1 Data access and upload

For catches within each country's exclusive economic zones, data are downloaded download data from the Sea Around Us site(Pauly, Zeller, and Palomares, 2020.). In the Biodiversity by Taxon Tools and Data, we filter by EEZ for All regions, and then filter by Taxon at the Species level for Common dolphinfish (Coryphaena hippurus). After clicking "View graphs" one can access the full data set using the "Download Data" button. The high seas data are accessed in the Basic Search High seas section of the site. The individual regions can be clicked on (e.g., Atlantic, Western Central) and then the data are dowloaded using the Download Data button.

The current version of data in use is version 50.1, accessed June 2024 (the data files can be found on github in the SEFSC-dolphin-analyses/data /SAU/ folder). This was still the most recent version of data available as of June 2025.

3.2 Read in the high seas landings

First we read in the high seas landings for the Western Central Atlantic and Northwest Atlantic zones. These are equivalent to FAO areas 21 and 31.

```
# clear workspace
rm(list = ls())

if(!require("pals")) install.packages("pals")
library(pals)

# find data files
hslis <- dir("data/SAU/")[grep("HighSeas", dir("data/SAU/"))]

# identify areas 21 and 31
hslis[1]; hslis[3]
```

```
[1] "SAU HighSeas 21 v50-1.csv"
```

```
[1] "SAU HighSeas 31 v50-1.csv"
```

```
#download data
h1 <- read.csv(paste0("data/SAU/", hslis[1]))
h3 <- read.csv(paste0("data/SAU/", hslis[3]))

# confirm correct areas
#unique(h1$area_name)
#unique(h3$area_name)

hs <- rbind(h1, h3)
```

We extract the data for dolphinfish and look at the characteristics of the landings. Most of the landings are in the Western Central Atlantic, and the primary fishing entities are Venezuela and Saint Lucia. The fishing sector is all industrial, and the vast majority of the catch is reported landings, with only a tiny fraction of discards or unreported landings listed in the database. The gear type is primarily long line.

```
# check species name and filter out dolphinfish
unique(hs$common_name)[grep("dolphin", unique(hs$common_name))]
```

```
[1] "Common dolphinfish"
```

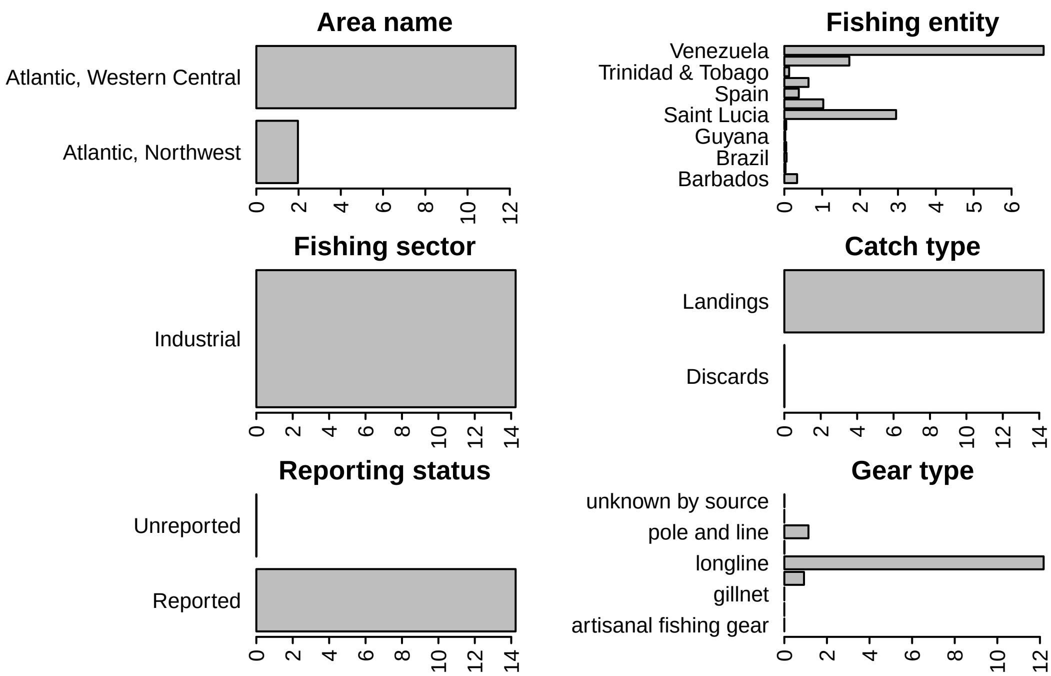
```r
hs <- hs[which(hs$common_name == "Common dolphinfish"), ]

# look at the categories within each column of data
apply(hs[1:13], 2, table)
```

```
$area_name

      Atlantic, Northwest Atlantic, Western Central
                      113                       160

$area_type

high_seas
      273

$year

1985 1986 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012
   1    1    1    1    1    1    1    6    4    1    5   10    5    8   10   12
2013 2014 2015 2016 2017 2018 2019
  24   22   30   33   33   29   34

$scientific_name

Coryphaena hippurus
                273

$common_name

Common dolphinfish
               273

$functional_group

Large pelagics (>=90 cm)
                     273

$commercial_group

Perch-likes
```

```
        273

$fishing_entity

                    Barbados                       Belize
                           6                            3
                      Brazil                       Canada
                          19                           18
                      Guyana                       Panama
                           1                            1
                 Saint Lucia Saint Vincent & the Grenadines
                           5                           28
                       Spain                     Suriname
                          80                            1
           Trinidad & Tobago                          USA
                          14                           74
                   Venezuela
                          23

$fishing_sector

Industrial
       273

$catch_type

Discards Landings
       5      268

$reporting_status

  Reported Unreported
       253         20

$gear_type

artisanal fishing gear          bottom trawl                gillnet
                     7                     9                     11
            hand lines              longline             mixed gear
                    17                   187                      1
         pole and line         pots or traps      unknown by source
```

```
                    19                     9                     13

$end_use_type

                         Direct human consumption    Fishmeal and fish oil
                       5                      194                      37
                   Other
                      37
```

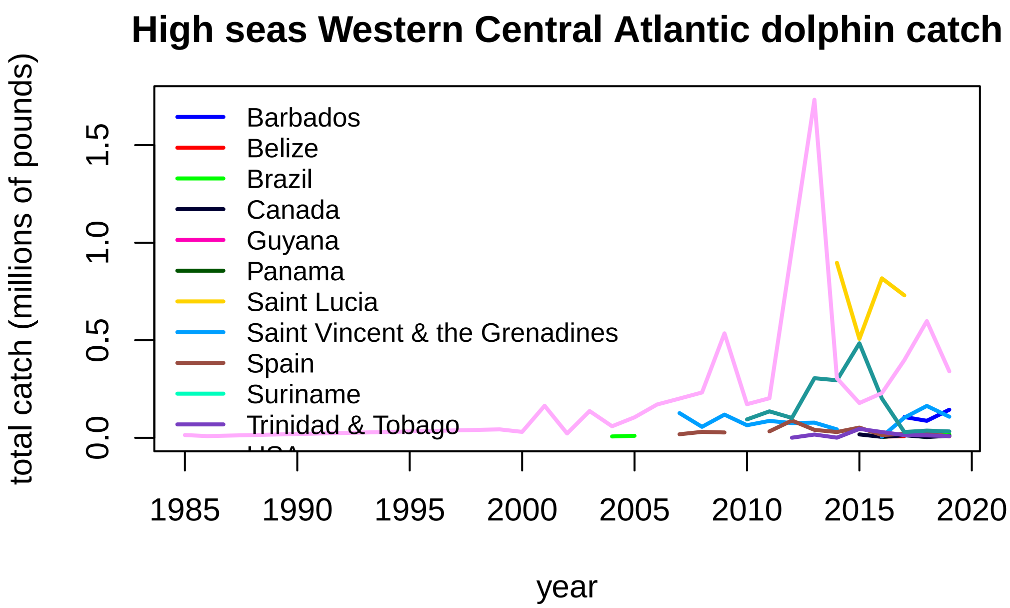
```
# convert from metric tonnes to pounds
hs$pounds <- hs$tonnes * 1000 * 2.20462

# produce barplot of key data fields
par(mar = c(3, 20, 3, 1), mfrow = c(3, 2), mex = 0.5)
barplot(tapply(hs$pounds/10^6, hs$area_name, sum, na.rm = T), las = 2, xlab = "millions of
#barplot(tapply(hs$pounds/10^6, hs$area_type, sum, na.rm = T), las = 2, ylab = "millions of
#barplot(tapply(hs$pounds/10^6, hs$year, sum, na.rm = T), las = 2, ylab = "millions of poun
barplot(tapply(hs$pounds/10^6, hs$fishing_entity, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(hs$pounds/10^6, hs$fishing_sector, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(hs$pounds/10^6, hs$catch_type, sum, na.rm = T), las = 2, xlab = "millions of
barplot(tapply(hs$pounds/10^6, hs$reporting_status, sum, na.rm = T), las = 2, xlab = "milli
barplot(tapply(hs$pounds/10^6, hs$gear_type, sum, na.rm = T), las = 2, xlab = "millions of
```



3.3 Summarize the high seas landings

We plot the annual catches by fishing entity to look at trends over time.

```
hs$fishing_entity <- as.factor(hs$fishing_entity)
hs$fishing_entity <- droplevels(hs$fishing_entity)

par(mar = c(5, 5, 3, 1), mfrow = c(1, 1))
tab <- tapply(hs$pounds, list(hs$year, hs$fishing_entity), sum, na.rm = T)
matplot(as.numeric(rownames(tab)), tab/10^6, type = "l", col = glasbey(ncol(tab)), lty = 1,
        main = "High seas Western Central Atlantic dolphin catch")
legend("topleft", colnames(tab), col = glasbey(ncol(tab)), lty = 1, lwd = 2, cex = 0.8, bty
```



Finally we output the subset of high seas dolphin catch for the two areas. Because we have already accounted for the USA landings in our national reporting and are aiming to only summarize international landings here, we filter out the USA landings from the Sea Around Us database.

```
hs <- hs[which(hs$fishing_entity != "USA"), ]

hs$year <- factor(hs$year, levels = 1950:2019)
tab <- tapply(hs$pounds, list(hs$year, hs$area_name), sum, na.rm = T)
tab[is.na(tab)] <- 0
tail(tab, 40)
```

```
     Atlantic, Northwest Atlantic, Western Central
1980                0.000                     0.000
1981                0.000                     0.000
1982                0.000                     0.000
1983                0.000                     0.000
1984                0.000                     0.000
1985                0.000                 14037.129
1986                0.000                  8748.399
1987                0.000                     0.000
```

```
1988            0.000            0.000
1989            0.000            0.000
1990            0.000            0.000
1991            0.000            0.000
1992            0.000            0.000
1993            0.000            0.000
1994            0.000            0.000
1995            0.000            0.000
1996            0.000            0.000
1997            0.000            0.000
1998            0.000            0.000
1999            0.000        43168.296
2000            0.000        30284.627
2001            0.000       164272.223
2002            0.000        22147.477
2003            0.000       137103.181
2004            0.000        66058.258
2005            0.000       115171.863
2006            0.000       170522.101
2007        14687.527       330838.620
2008        28552.636       340863.719
2009        18719.121       662720.661
2010         4510.085       232093.837
2011        21589.424       300498.224
2012        72398.812      1059210.392
2013        34741.604      2471791.396
2014        27922.953      1247462.983
2015        90313.638       710059.125
2016        15319.433      1094893.891
2017        34806.933      1374259.739
2018        29328.358       885814.754
2019        21870.621       632747.157
```

```r
# write summary to merge with EEZ data
write.csv(tab, file = "data/SAU/high_seas_by_year.csv")
```

3.4 Read in the landings within excluzive economic zones (EEZs)

Now we read in the landings reported by each country or jurisdiction's respective EEZ. Because we have extracted dolphin landings for EEZs worldwide, we need to separate out the landings for the Atlantic Ocean basin.

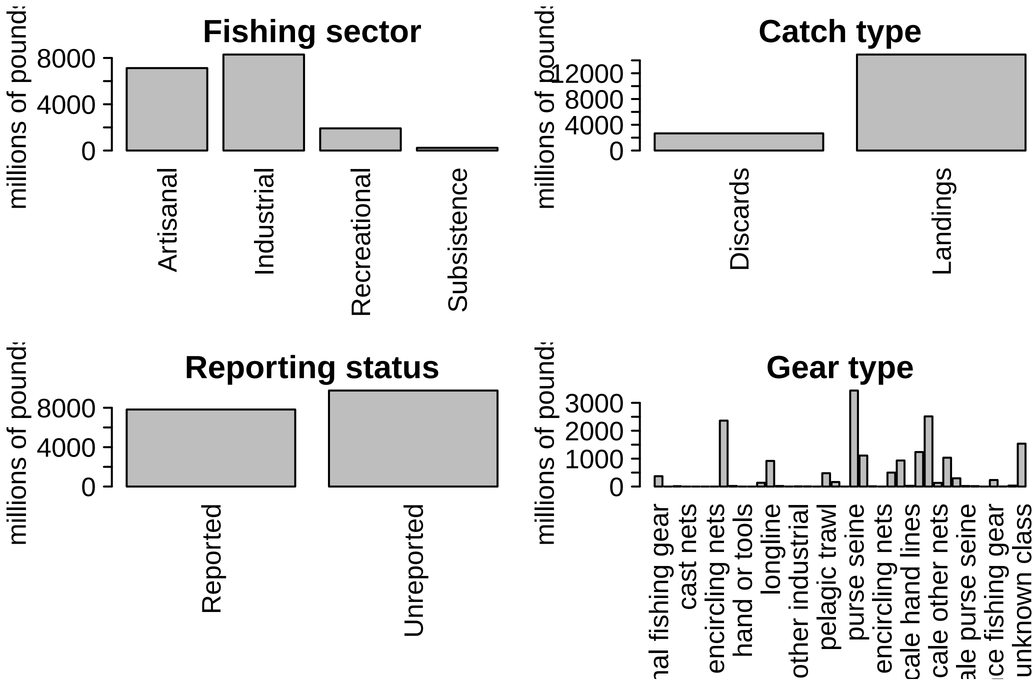
```
rm(list = ls())
d <- read.csv("data/SAU/SAU Taxa 600006 v50-1.csv", header = T, sep = ",")  # v. 50.1 - Jun
head(d)
```

```
     area_name area_type year     scientific_name        common_name
1    Australia       eez 1950 Coryphaena hippurus Common dolphinfish
2    Australia       eez 1950 Coryphaena hippurus Common dolphinfish
3    Australia       eez 1950 Coryphaena hippurus Common dolphinfish
4      Bahamas       eez 1950 Coryphaena hippurus Common dolphinfish
5     Barbados       eez 1950 Coryphaena hippurus Common dolphinfish
6 Bermuda (UK)       eez 1950 Coryphaena hippurus Common dolphinfish
          functional_group commercial_group fishing_entity fishing_sector
1 Large pelagics (>=90 cm)      Perch-likes      Australia     Industrial
2 Large pelagics (>=90 cm)      Perch-likes      Australia     Industrial
3 Large pelagics (>=90 cm)      Perch-likes      Australia     Industrial
4 Large pelagics (>=90 cm)      Perch-likes        Bahamas   Recreational
5 Large pelagics (>=90 cm)      Perch-likes       Barbados   Recreational
6 Large pelagics (>=90 cm)      Perch-likes   Bermuda (UK)      Artisanal
  catch_type reporting_status                 gear_type
1   Landings         Reported                  longline
2   Landings         Reported                  longline
3   Landings         Reported                  longline
4   Landings       Unreported recreational fishing gear
5   Landings       Unreported recreational fishing gear
6   Landings         Reported         small scale lines
             end_use_type       tonnes landed_value
1    Fishmeal and fish oil  0.003373202 3.438170e-05
2 Direct human consumption  6.739657104 1.040254e+04
3                    Other  0.003373202 3.438170e-05
4 Direct human consumption 48.665105143 8.262728e+00
5 Direct human consumption  0.009158559 4.846886e-04
6 Direct human consumption  1.511085514 3.199403e+03
```

```
#dim(d)
#apply(d[1:13], 2, table)

# convert tonnes to pounds
d$lbs <- d$tonnes * 1000 * 2.20462
```

We look at the characteristics of the landings within EEZs. These are largely industrial and artisanal in nature. Landings are the majority of the catch but discards are also present. There are more unreported landings than reported landings and a variety of gear types are used.

```
# look at characteristics of data
par(mfrow = c(2, 2), mar = c(12, 7, 3, 1), mex = 0.5, mgp = c(5, 1, 0))
barplot(tapply(d$lbs, d$fishing_sector, sum, na.rm = T)/10^6, las = 2, ylab = "millions of
barplot(tapply(d$lbs, d$catch_type, sum, na.rm = T)/10^6, las = 2, ylab = "millions of poun
barplot(tapply(d$lbs, d$reporting_status, sum, na.rm = T)/10^6, las = 2, ylab = "millions o
barplot(tapply(d$lbs, d$gear_type, sum, na.rm = T)/10^6, las = 2, ylab = "millions of pound
```



3.5 Separating the EEZs by area

Now we have to manually assign the different EEZs to respective areas. We separate the geographical areas within the Atlantic Ocean (North and South) and lump all other EEZs into "other oceans." This latter category is largely the Pacific Ocean and to a lesser extent the Indian Ocean. We remove the high seas catch from this database as it is not ocean-specific and we have extracted the ocean-specific high seas catch previously.

```
# remove the high seas catch from this database
d <- d[which(d$area_type != "high_seas"), ]
d$fishing_entity <- as.factor(d$fishing_entity)
d$fishing_entity <- droplevels(d$fishing_entity)
table(d$area_type)  # this should be all "eez"
```

```
  eez
63056
```

```
# Western Central Atlantic EEZs
WCA <- c("Bahamas", "Barbados", "Cayman Isl. (UK)", "Dominica", "Dominican Republic", "Gren
         "Aruba (Netherlands)", "Haiti", "Guadeloupe  (France)", "Jamaica", "Martinique (Fr
         "Montserrat (UK)", "Nicaragua (Caribbean)", "Puerto Rico (USA)", "Saint Kitts & Ne
         "Saint Lucia", "Saint Vincent & the Grenadines", "Trinidad & Tobago", "US Virgin I
         "St Barthelemy (France)", "St Martin (France)", "Curaçao (Netherlands)", "Bonaire 
         "Saba and Sint Eustatius (Netherlands)", "Sint Maarten (Netherlands)", "Honduras (
         "Colombia (Caribbean)", "Mexico (Atlantic)", "Turks & Caicos Isl. (UK)", "Guyana",
         "Antigua & Barbuda", "Belize", "British Virgin Isl. (UK)", "French Guiana", "Cuba"
         "Anguilla (UK)", "Suriname", "Costa Rica (Caribbean)", "Guatemala (Caribbean)",
         "Panama (Caribbean)", "Bermuda (UK)", "CuraÃ§ao (Netherlands)")

# Brazil
Bra <- c("Brazil (mainland)", "Fernando de Noronha (Brazil)",
         "St Paul and St. Peter Archipelago (Brazil)")

# Canada
Can <-  c("Canada (East Coast)", "Saint Pierre & Miquelon (France)")

# United States
USA <-  c("USA (East Coast)", "USA (Gulf of Mexico)")
```

```
# Northeast Atlantic EEZs
NEA <- c("Canary Isl. (Spain)", "Spain (mainland; Med and Gulf of Cadiz)", "Spain (Northwes
         "Spain (mainland, Med and Gulf of Cadiz)",
         "France (Atlantic Coast)", "Portugal (mainland)", "Azores Isl. (Portugal)")

# Eastern Central Atlantic EEZs
ECA <- c("Madeira Isl. (Portugal)", "Cape Verde", "Equatorial Guinea", "Benin",
          "Sao Tome & Principe ", "Sao Tome & Principe", "Morocco (Central)", "Ghana",
          "Côte d'Ivoire", "CÃ´te d'Ivoire", "Guinea", "Liberia", "Sierra Leone", "Nigeria"
          "Guinea-Bissau", "Senegal", "Cameroon", "Gambia", "Mauritania", "Morocco (South)"
          "Gabon", "Congo; R. of", "Congo, R. of", "Congo (ex-Zaire)")

# Southeastern Atlantic EEZs
SEA <- c("Angola", "Namibia", "South Africa (Atlantic and Cape)",
           "Ascension Isl. (UK)", "Saint Helena (UK)", "Tristan da Cunha Isl. (UK)")

# Southwest Atlantic EEZs
SWA <- c("Trindade & Martim Vaz Isl. (Brazil)", "Uruguay")

# Mediterranean EEZs
Med <- c("Italy (mainland)", "Libya", "Malta", "Syria", "Sicily (Italy)","Sardinia (Italy)"
         "Balearic Islands (Spain)", "Cyprus (North)", "Cyprus (South)",
         "Gaza Strip", "Greece (without Crete)", "Crete (Greece)", "Lebanon",
         "Turkey (Mediterranean Sea)", "Egypt (Mediterranean)", "Israel (Mediterranean)", "
         "Algeria", "Albania", "Croatia", "Montenegro", "Morocco (Mediterranean)",
         "France (Mediterranean)", "Corsica (France)")

# add variable to specify region
d$reg <- "other oceans"
d$reg[which(d$area_name %in% WCA)] <- "Western Central Atlantic EEZs"
d$reg[which(d$area_name %in% Bra)] <- "Brazil EEZ"
d$reg[which(d$area_name %in% Can)] <- "Canadian EEZ"
d$reg[which(d$area_name %in% USA)] <- "United States EEZ"
d$reg[which(d$area_name %in% NEA)] <- "Northeast Atlantic EEZs"
d$reg[which(d$area_name %in% ECA)] <- "Eastern Central Atlantic EEZs"
d$reg[which(d$area_name %in% SEA)] <- "Southeast Atlantic EEZs"
d$reg[which(d$area_name %in% SWA)] <- "Southwest Atlantic EEZs"
d$reg[which(d$area_name %in% Med)] <- "Mediterranean Sea EEZs"
```

Now we will check that we have classified the regions correctly.

```r
# check region designation
othlis <- unique(d$area_name[which(d$reg == "other oceans")])
print("EEZs of all other oceans: ")
```

```
[1] "EEZs of all other oceans: "
```

```r
othlis
```

```
 [1] "Australia"                         "China"
 [3] "Taiwan"                            "Mayotte (France)"
 [5] "French Polynesia"                  "Hong Kong (China)"
 [7] "Andaman & Nicobar Isl. (India)"    "Indonesia (Eastern)"
 [9] "Japan (main islands)"              "Japan (Daito Islands)"
[11] "Korea (South)"                     "Hawaii Northwest Islands (USA)"
[13] "Nauru"                             "Niue (New Zealand)"
[15] "Northern Marianas (USA)"           "Micronesia (Federated States of)"
[17] "Marshall Isl."                     "Palau"
[19] "Pakistan"                          "Papua New Guinea"
[21] "Philippines"                       "Seychelles"
[23] "Viet Nam"                          "Tokelau (New Zealand)"
[25] "Tonga"                             "Hawaii Main Islands (USA)"
[27] "USA (West Coast)"                  "Wake Isl. (USA)"
[29] "Wallis & Futuna Isl. (France)"     "Samoa"
[31] "Guatemala (Pacific)"               "Kiribati (Gilbert Islands)"
[33] "Mexico (Pacific)"                  "Japan (Ogasawara Islands)"
[35] "Korea (North, Yellow Sea)"         "Korea (North, Sea of Japan)"
[37] "Mauritius"                         "Solomon Isl."
[39] "Christmas Isl. (Australia)"        "Johnston Atoll (USA)"
[41] "Timor Leste"                       "Palmyra Atoll & Kingman Reef (USA)"
[43] "Jarvis Isl. (USA)"                 "Howland & Baker Isl. (USA)"
[45] "Indonesia (Central)"               "Indonesia (Indian Ocean)"
[47] "Kiribati (Line Islands)"           "Kiribati (Phoenix Islands)"
[49] "New Caledonia (France)"            "Vanuatu"
[51] "Lord Howe Isl. (Australia)"        "Myanmar"
[53] "Ecuador (mainland)"                "El Salvador"
[55] "Fiji"                              "Malaysia (Peninsula West)"
[57] "Norfolk Isl. (Australia)"          "Peru"
[59] "Pitcairn (UK)"                     "Colombia (Pacific)"
[61] "Costa Rica (Pacific)"              "Nicaragua (Pacific)"
```

```
 [63] "Panama (Pacific)"                     "Thailand (Andaman Sea)"
 [65] "Thailand (Gulf of Thailand)"          "Cook Islands"
 [67] "Kermadec Isl. (New Zealand)"          "Tuvalu"
 [69] "Honduras (Pacific)"                   "American Samoa"
 [71] "Galapagos Isl. (Ecuador)"             "New Zealand"
 [73] "Brunei Darussalam"                    "Malaysia (Sabah)"
 [75] "Malaysia (Sarawak)"                   "Guam (USA)"
 [77] "Clipperton Isl. (France)"             "South Africa (Indian Ocean Coast)"
 [79] "Singapore"                            "Easter Isl. (Chile)"
 [81] "Madagascar"                           "Réunion (France)"
 [83] "Chagos Archipelago (UK)"              "Comoros Isl."
 [85] "Mozambique Channel Isl. (France)"     "Kenya"
 [87] "Mozambique"                           "Somalia"
 [89] "Tanzania"                             "Glorieuse Islands (France)"
 [91] "Yemen (Socotra)"                      "Oman"
 [93] "St Paul & Amsterdam Isl. (France)"    "Iran (Sea of Oman)"
 [95] "Desventuradas Isl. (Chile)"           "India (mainland)"
 [97] "Maldives"                             "Tromelin Isl. (France)"
 [99] "Cambodia"                             "Sri Lanka"
[101] "Malaysia (Peninsula East)"            "Bangladesh"
```

```
print("U.S. EEZs outside the Atlantic Ocean: ")
```

```
[1] "U.S. EEZs outside the Atlantic Ocean: "
```

```
othlis[grep("USA", othlis)]
```

```
 [1] "Hawaii Northwest Islands (USA)"     "Northern Marianas (USA)"
 [3] "Hawaii Main Islands (USA)"          "USA (West Coast)"
 [5] "Wake Isl. (USA)"                    "Johnston Atoll (USA)"
 [7] "Palmyra Atoll & Kingman Reef (USA)" "Jarvis Isl. (USA)"
 [9] "Howland & Baker Isl. (USA)"         "Guam (USA)"
```

```
print("French EEZs outside the Atlantic Ocean: ")
```

```
[1] "French EEZs outside the Atlantic Ocean: "
```

```r
othlis[grep("France", othlis)]
```

```
[1] "Mayotte (France)"                "Wallis & Futuna Isl. (France)"
[3] "New Caledonia (France)"          "Clipperton Isl. (France)"
[5] "Réunion (France)"                "Mozambique Channel Isl. (France)"
[7] "Glorieuse Islands (France)"      "St Paul & Amsterdam Isl. (France)"
[9] "Tromelin Isl. (France)"
```

```r
unique(d$area_name[which(d$reg == "Western Central Atlantic EEZs")])   # Western Central EE
```

```
 [1] "Bahamas"
 [2] "Barbados"
 [3] "Bermuda (UK)"
 [4] "Cayman Isl. (UK)"
 [5] "Dominica"
 [6] "Dominican Republic"
 [7] "Grenada"
 [8] "Guadeloupe  (France)"
 [9] "Haiti"
[10] "Jamaica"
[11] "Martinique (France)"
[12] "Montserrat (UK)"
[13] "Puerto Rico (USA)"
[14] "Saint Kitts & Nevis"
[15] "Saint Lucia"
[16] "Saint Vincent & the Grenadines"
[17] "Trinidad & Tobago"
[18] "US Virgin Isl."
[19] "St Barthelemy (France)"
[20] "St Martin (France)"
[21] "Curaçao (Netherlands)"
[22] "Bonaire (Netherlands)"
[23] "Saba and Sint Eustatius (Netherlands)"
[24] "Sint Maarten (Netherlands)"
[25] "Honduras (Caribbean)"
[26] "Colombia (Caribbean)"
[27] "Mexico (Atlantic)"
[28] "Turks & Caicos Isl. (UK)"
[29] "Costa Rica (Caribbean)"
```

```
[30] "Panama (Caribbean)"
[31] "Antigua & Barbuda"
[32] "Belize"
[33] "British Virgin Isl. (UK)"
[34] "Cuba"
[35] "French Guiana"
[36] "Guyana"
[37] "Anguilla (UK)"
[38] "Suriname"
[39] "Venezuela"
[40] "Guatemala (Caribbean)"
[41] "Aruba (Netherlands)"
[42] "Nicaragua (Caribbean)"
```

```
unique(d$area_name[which(d$reg == "Brazil EEZ")])   # Brazil EEZs:
```

```
[1] "Brazil (mainland)"
[2] "Fernando de Noronha (Brazil)"
[3] "St Paul and St. Peter Archipelago (Brazil)"
```

```
unique(d$area_name[which(d$reg == "Canadian EEZ")])   # Canadian EEZs:
```

```
[1] "Canada (East Coast)"              "Saint Pierre & Miquelon (France)"
```

```
unique(d$area_name[which(d$reg == "United States EEZ")])  # United States EEZs:
```

```
[1] "USA (East Coast)"     "USA (Gulf of Mexico)"
```

```
unique(d$area_name[which(d$reg == "Northeast Atlantic EEZs")])   # Northeast Atlantic EEZs:
```

```
[1] "Azores Isl. (Portugal)"
[2] "Spain (mainland, Med and Gulf of Cadiz)"
[3] "Canary Isl. (Spain)"
[4] "Portugal (mainland)"
[5] "France (Atlantic Coast)"
[6] "Spain (Northwest)"
```

```
unique(d$area_name[which(d$reg == "Eastern Central Atlantic EEZs")])   # Eastern Central At
```

```
 [1] "Cape Verde"                 "Equatorial Guinea"
 [3] "Sao Tome & Principe"        "Madeira Isl. (Portugal)"
 [5] "Togo"                       "Ghana"
 [7] "Guinea"                     "Liberia"
 [9] "Guinea-Bissau"              "Senegal"
[11] "Sierra Leone"               "Morocco (Central)"
[13] "Morocco (South)"            "Côte d'Ivoire"
[15] "Gabon"                      "Benin"
[17] "Gambia"                     "Mauritania"
[19] "Nigeria"                    "Cameroon"
[21] "Congo, R. of"               "Congo (ex-Zaire)"
```

```
unique(d$area_name[which(d$reg == "Southeast Atlantic EEZs")])   # Southeast Atlantic EEZs:
```

```
[1] "Ascension Isl. (UK)"                "South Africa (Atlantic and Cape)"
[3] "Saint Helena (UK)"                  "Angola"
[5] "Namibia"                            "Tristan da Cunha Isl. (UK)"
```

```
unique(d$area_name[which(d$reg == "Southwest Atlantic EEZs")])   # Southwest Atlantic EEZs:
```

```
[1] "Trindade & Martim Vaz Isl. (Brazil)" "Uruguay"
```

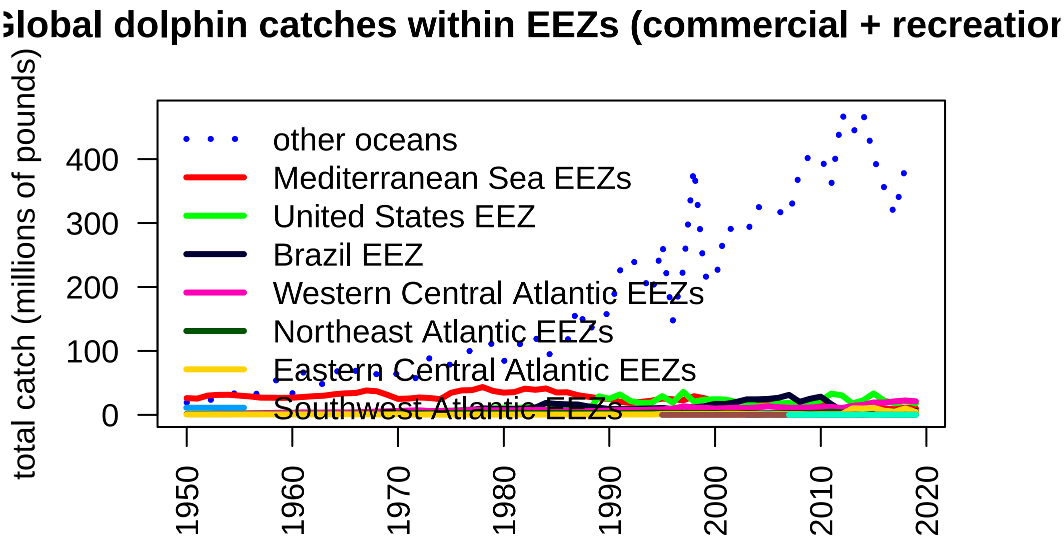
```
unique(d$area_name[which(d$reg == "Mediterranean Sea EEZs")])   # Mediterranean Sea EEZs:
```

```
 [1] "Italy (mainland)"          "Libya"
 [3] "Malta"                     "Syria"
 [5] "Sicily (Italy)"            "Sardinia (Italy)"
 [7] "Balearic Islands (Spain)" "Tunisia"
 [9] "Algeria"                   "Morocco (Mediterranean)"
[11] "France (Mediterranean)"    "Corsica (France)"
[13] "Croatia"                   "Lebanon"
[15] "Albania"                   "Greece (without Crete)"
[17] "Crete (Greece)"            "Cyprus (South)"
```

3.6 Summarize catch by regional EEZs

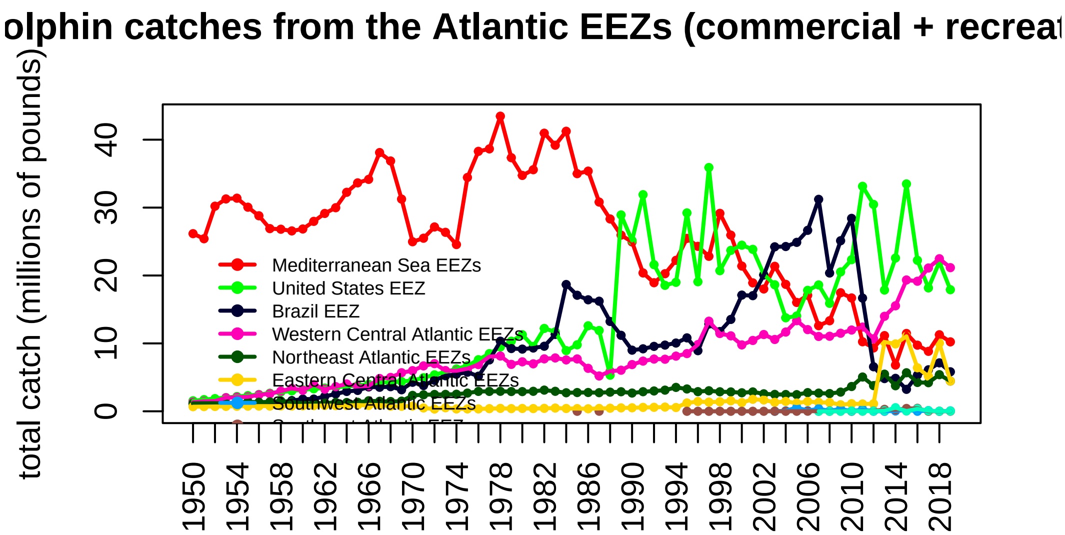
Now we will summarize the catch by region by grouping the catch across the EEZs by region and plot the results. The U.S. landings should be very close to what is reported in Section 2 because the majority of U.S. catch is reported through the various data collection systems that are compiled within Sea Around Us.

```
# summarize EEZ catch by region
tab <- tapply(d$lbs, list(d$year, d$reg), sum, na.rm = T)
#head(tab)
tab2 <- t(tab)
tab3 <- tab2[order(rowSums(tab2, na.rm = T), decreasing = T), ]
tab_glob <- t(tab3)
#head(tab_glob)
yrs <- as.numeric(rownames(tab))

par(mar = c(5, 4, 4, 3), mgp = c(3, 1, 0))
matplot(yrs, tab_glob/10^6, las = 2, type = "l", lty = c(3, rep(1, 9)), lwd = 3, col = glas
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Global dolphin catches within EEZs (commercial + recreational)")
legend("topleft", colnames(tab_glob), col = glasbey(ncol(tab_glob)),
       lty = c(3, rep(1, 9)), lwd = 3, bty = "n")
```



We can see that the EEZ catches globally are dominated by areas outside of the Atlantic ("other oceans") so we will replot the data after moving this category. Here we can see that most of the catches historically were coming from the Mediterranean Sea, though these catches have been declining drastially since the 1970s. Catches in the United States have showed an opposite pattern, with low historical catches that increased in the 1980s (largely due to the introduction of recreational monitoring). Catches in the Western Central EEZs have increased gradually over time.

```
# remove other oceans
#head(tab_glob)
tab <- tab_glob[, -which(colnames(tab_glob) == "other oceans")]
#head(tab)

matplot(yrs, tab/10^6, las = 2, type = "b", pch = 19, cex = 0.5, lty = 1, col = glasbey(nco
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Dolphin catches from the Atlantic EEZs (commercial + recreational)")
matplot(yrs, tab/10^6, las = 2, type = "l", pch = 19, lty = 1, lwd = 2, add = T, col = glas
axis(1, at = yrs[seq(1, 90, 2)], las = 2); axis(2); box()
legend(1952, 25, colnames(tab), col = glasbey(ncol(tab)+1)[-1], pch = 19, lty = 1, lwd = 2,
```



3.7 Merge high seas with catch with exclusive economic zones

Now we will merge in the high seas catch from the steps above into the EEZ catches to get a better picture of total catch in FAO zones 31 and 21. Recall that the U.S. Gulf and Atlantic catches are not necessarily parsed apart correctly in the public databases which get compiled into Sea Around Us, so these will have the same deficiencies as noted in Section 2. Later we will merge in the corrected data.

```
hs <- read.csv("data/SAU/high_seas_by_year.csv")
names(hs)[1] <- "year"

# check that columns match
#head(tab)
#head(hs)
table(rownames(tab) == hs$year)
```

```
TRUE
  70
```

```
tab[which(is.na(tab))] <- 0
tab_atl <- cbind(tab, hs[2:3])

# check the merge
head(tab_atl)
```

```
     Mediterranean Sea EEZs United States EEZ Brazil EEZ
1950               26162262           1548245    1127242
1951               25410830           1721403    1060734
1952               30206350           1871779    1128992
1953               31275619           2024535    1111008
1954               31383840           2177865    1090492
1955               30056210           2309815    1190424
     Western Central Atlantic EEZs Northeast Atlantic EEZs
1950                       1345027                 1173441
1951                       1467925                 1187386
1952                       1510406                 1193045
1953                       2001465                 1141504
1954                       2069754                 1269523
1955                       2130341                 1405543
     Eastern Central Atlantic EEZs Southwest Atlantic EEZs
1950                      677552.5                       0
1951                      693004.4                       0
1952                      708466.7                       0
1953                      723939.4                       0
1954                      739422.5                       0
1955                      754915.9                       0
     Southeast Atlantic EEZs Canadian EEZ Atlantic..Northwest
1950                       0            0                   0
1951                       0            0                   0
1952                       0            0                   0
1953                       0            0                   0
1954                       0            0                   0
1955                       0            0                   0
     Atlantic..Western.Central
1950                         0
1951                         0
1952                         0
1953                         0
1954                         0
```

```
1955                         0
```

```
# separate all WCA catches
unique(d$reg)  # use Western Central, U.S. and Canada
```

```
 [1] "other oceans"                    "Western Central Atlantic EEZs"
 [3] "Brazil EEZ"                      "Eastern Central Atlantic EEZs"
 [5] "Mediterranean Sea EEZs"          "Northeast Atlantic EEZs"
 [7] "United States EEZ"               "Southeast Atlantic EEZs"
 [9] "Southwest Atlantic EEZs"         "Canadian EEZ"
```

```
dwc <- d[which(d$reg == "Western Central Atlantic EEZs" | d$reg == "United States EEZ" | d$

#unique(dwc$area_name[which(dwc$reg == "United States EEZ")])
#unique(dwc$area_name[which(dwc$reg == "Canadian EEZ")])
#unique(dwc$area_name[which(dwc$reg == "Western Central Atlantic EEZs")])

# relabel U.S. by area
dwc$reg[which(dwc$area_name == "USA (East Coast)")] <- "United States East Coast"
dwc$reg[which(dwc$area_name == "USA (Gulf of Mexico)")] <- "United States Gulf"
dwc$reg[which(dwc$area_name == "Puerto Rico (USA)")] <- "United States Caribbean"
dwc$reg[which(dwc$area_name == "US Virgin Isl.")] <- "United States Caribbean"
#table(dwc$area_name, dwc$reg)

tab_wc <- tapply(dwc$lbs, list(dwc$year, dwc$reg), sum, na.rm = T)
#head(tab_wc)

# reorder S to N
tab_wc <- tab_wc[, c(5, 2, 4, 3, 1)]
head(tab_wc)
```

```
     Western Central Atlantic EEZs United States Caribbean United States Gulf
1950                       1202464                142562.8           309225.3
1951                       1299559                168365.6           339921.3
1952                       1341943                168463.5           370512.4
1953                       1839330                162135.2           400578.5
1954                       1901095                168659.3           433479.6
1955                       1954511                175830.1           463752.0
     United States East Coast Canadian EEZ
```

```
1950                  1239020            NA
1951                  1381482            NA
1952                  1501267            NA
1953                  1623956            NA
1954                  1744386            NA
1955                  1846063            NA
```

```
# merge in the high seas catch - check years match
#head(hs)
table(rownames(tab_wc) == hs$year)
```

```
TRUE
  70
```

```
high_seas <- hs$Atlantic..Western.Central + hs$Atlantic..Northwest
high_seas[high_seas == 0] <- NA
tab_wc <- cbind(tab_wc, high_seas)
head(tab_wc)
```

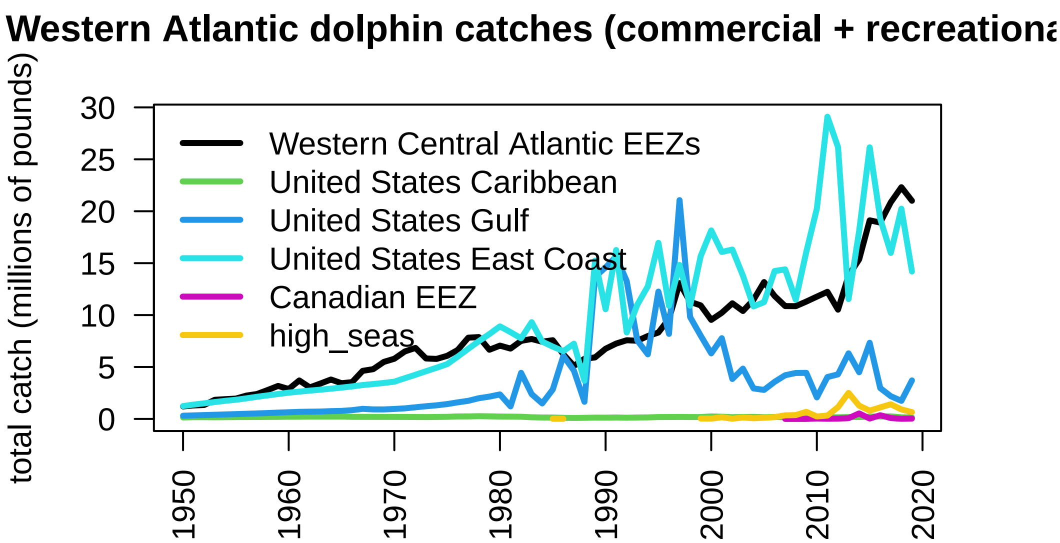
```
     Western Central Atlantic EEZs United States Caribbean United States Gulf
1950                       1202464                142562.8           309225.3
1951                       1299559                168365.6           339921.3
1952                       1341943                168463.5           370512.4
1953                       1839330                162135.2           400578.5
1954                       1901095                168659.3           433479.6
1955                       1954511                175830.1           463752.0
     United States East Coast Canadian EEZ high_seas
1950                  1239020           NA        NA
1951                  1381482           NA        NA
1952                  1501267           NA        NA
1953                  1623956           NA        NA
1954                  1744386           NA        NA
1955                  1846063           NA        NA
```

3.8 Plot Western Atlantic catches

Now we plot the total Western Atlantic catches – both within the EEZs and the high seas – separating by U.S., Canada, and the Western Central Atlantic (i.e., the Wider

Caribbean). Most of the catch is coming from the United States and the 40 combined Caribbean jurisdictions which comprise the Western Central Atlantic.

```
par(mar = c(5, 4, 4, 3), mgp = c(3, 1, 0))
matplot(yrs, tab_wc/10^6, las = 2, type = "l", lty = 1, lwd = 3, col = c(1, 3:7),
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Western Atlantic dolphin catches (commercial + recreational)")
legend("topleft", colnames(tab_wc), col = c(1, 3:7), lty = 1, lwd = 3, bty = "n")
```



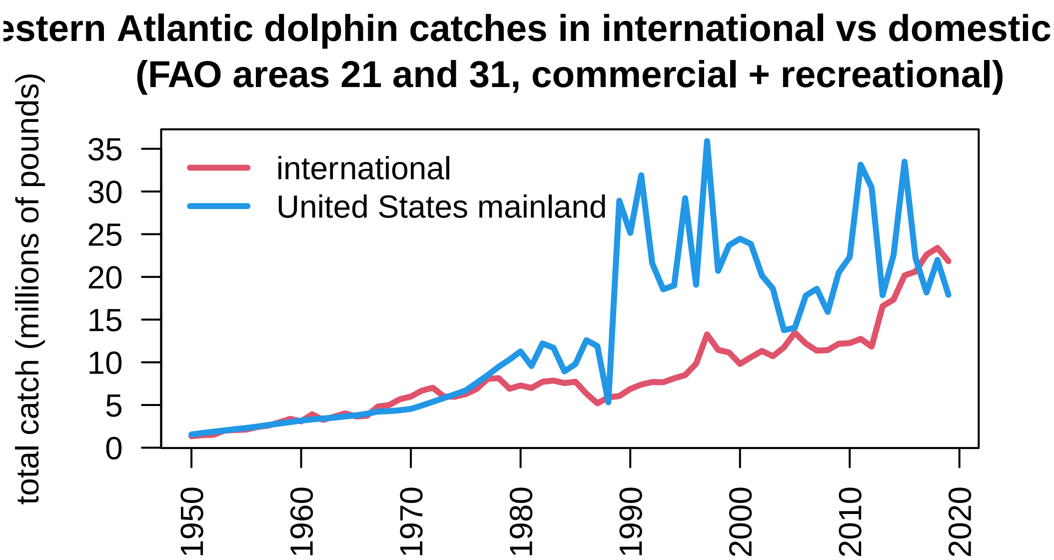
Separating out the mainland United States catches, here are the international and domestic (mainland) catches plotted over time.

```
tab_int <- tab_wc[, c(1, 2, 5, 6)]
head(tab_int)
```

```
     Western Central Atlantic EEZs United States Caribbean Canadian EEZ
1950                       1202464                142562.8           NA
1951                       1299559                168365.6           NA
1952                       1341943                168463.5           NA
1953                       1839330                162135.2           NA
1954                       1901095                168659.3           NA
```

```
1955                         1954511                175830.1           NA
     high_seas
1950         NA
1951         NA
1952         NA
1953         NA
1954         NA
1955         NA
```

```
int_catch <- round(rowSums(tab_int, na.rm = T), 2)
dom_catch <- rowSums(tab_wc[, c(3, 4)], na.rm = T)

matplot(yrs, cbind(int_catch, dom_catch)/10^6, las = 2, type = "l", lty = 1, lwd = 3, col =
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Western Atlantic dolphin catches in international vs domestic waters\n(FAO
legend("topleft", c("international", "United States mainland"), col = c(2, 4), lty = 1, lwd
```

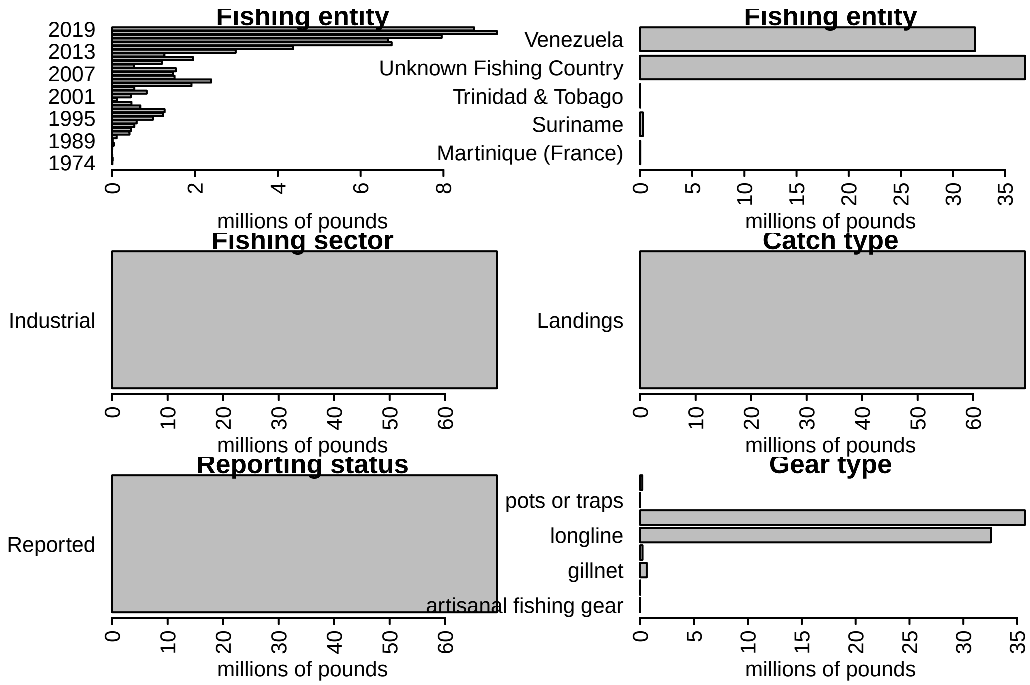



```
# output international catches
write.csv(data.frame(int_catch), file = "data/international_catches.csv", row.names = T)
```

3.9 Look at Venezuela catches in detail

Note in the plot above a rapid increase in international catches within the Western Atlantic. This is driven by a sudden increase in catches reported within Sea Around Us within the EEZ. We investigate these catches in detail as it seems odd that a relatively small fleet would be able to ramp up effort so rapidly. Data can be downloaded on a country-specific basis, the catches by taxon in the waters of Venezuela are accessed here.

```
rm(list = ls())  # clear the workspace

# download the data for waters of Venezuela
ven <- read.csv("data/SAU/SAU EEZ 862 v50-1.csv")

# subset dolphinfish from the data
v <- ven[which(ven$common_name == "Common dolphinfish"), ]

#apply(v[1:13], 2, table)

#convert to pounds
v$pounds <- v$tonnes * 1000 * 2.20462

# look at the characteristics of the data
par(mar = c(5, 10, 1, 1), mfrow = c(3, 2), mex = 0.5)
#barplot(tapply(v$pounds/10^6, v$area_name, sum, na.rm = T), las = 2, xlab = "millions of p
#barplot(tapply(v$pounds/10^6, v$area_type, sum, na.rm = T), las = 2, xlab = "millions of p
barplot(tapply(v$pounds/10^6, v$year, sum, na.rm = T), las = 2, xlab = "millions of pounds"
barplot(tapply(v$pounds/10^6, v$fishing_entity, sum, na.rm = T), las = 2, xlab = "millions
barplot(tapply(v$pounds/10^6, v$fishing_sector, sum, na.rm = T), las = 2, xlab = "millions
barplot(tapply(v$pounds/10^6, v$catch_type, sum, na.rm = T), las = 2, xlab = "millions of p
barplot(tapply(v$pounds/10^6, v$reporting_status, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(v$pounds/10^6, v$gear_type, sum, na.rm = T), las = 2, xlab = "millions of po
```



Note that a large portion of the catch in the database is listed as coming from an "Unknown Fishing Country." We extract these catches and further investigate the characteristics.

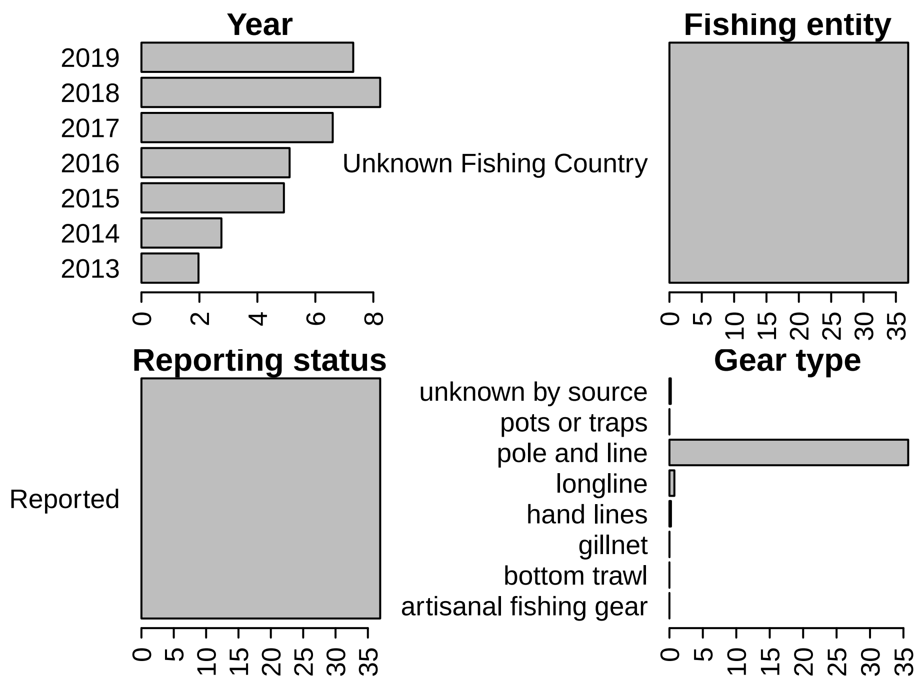
```
# extract the Unknown Fishing Country catch
v1 <- v[which(v$fishing_entity == "Unknown Fishing Country"), ]
apply(v1[1:15], 2, table)
```

```
$area_name

Venezuela
       39

$area_type

eez
 39

$data_layer

Assigned Tuna RFMO catch
```

```
                      39

$uncertainty_score
< table of extent 0 >

$year

2013 2014 2015 2016 2017 2018 2019
   5    5    5    6    7    5    6

$scientific_name

Coryphaena hippurus
                 39

$common_name

Common dolphinfish
                39

$functional_group

Large pelagics (>=90 cm)
                      39

$commercial_group

Perch-likes
         39

$fishing_entity

Unknown Fishing Country
                     39

$fishing_sector

Industrial
        39

$catch_type
```

```
Landings
      39

$reporting_status

Reported
      39

$gear_type

artisanal fishing gear           bottom trawl                gillnet
                     1                      5                      7
            hand lines               longline          pole and line
                     3                      4                      7
         pots or traps      unknown by source
                     5                      7

$end_use_type

Direct human consumption
                      39
```

```
# look at characteristics
par(mar = c(3, 10, 1, 5), mfrow = c(2, 2), mex = 0.6)
barplot(tapply(v1$pounds/10^6, v1$year, sum, na.rm = T), las = 2, xlab = "millions of pound
barplot(tapply(v1$pounds/10^6, v1$fishing_entity, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(v1$pounds/10^6, v1$reporting_status, sum, na.rm = T), las = 2, xlab = "milli
barplot(tapply(v1$pounds/10^6, v1$gear_type, sum, na.rm = T), las = 2, xlab = "millions of
```

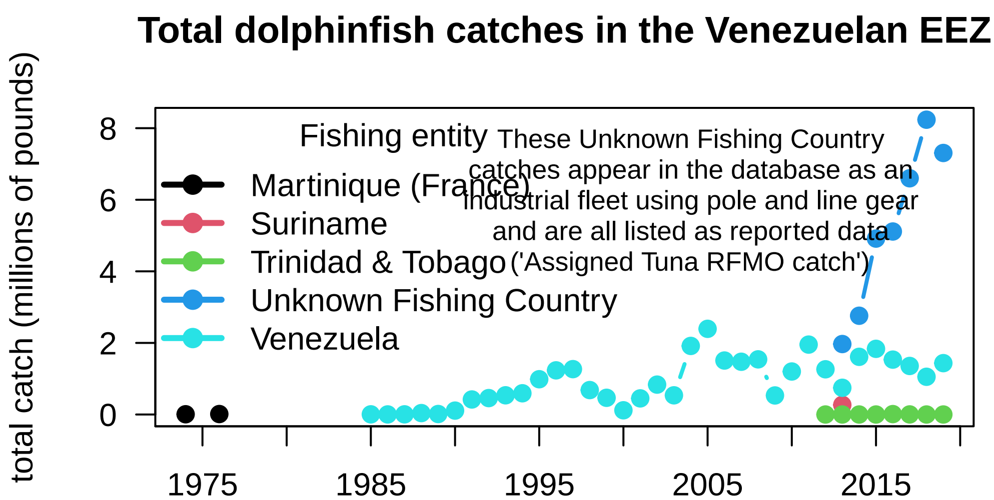


Here is another look at the catches within the venezuelan EEZ, separated by fishing entity.

```
# summarize Venezuela EEZ catches by fishing entity
tab <- tapply(v$pounds, list(v$year, v$fishing_entity), sum, na.rm = T)
head(tab)
```

```
     Martinique (France) Suriname Trinidad & Tobago Unknown Fishing Country
1974            7320.429       NA                NA                      NA
1976           12814.641       NA                NA                      NA
1985                  NA       NA                NA                      NA
1986                  NA       NA                NA                      NA
1987                  NA       NA                NA                      NA
1988                  NA       NA                NA                      NA
     Venezuela
1974        NA
1976        NA
1985  4699.563
1986  2229.484
1987  4585.610
1988 39960.149
```

```
matplot(rownames(tab), tab/10^6, las = 2, type = "b", lty = 1, lwd = 2, pch = 19, col = 1:5
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Total dolphinfish catches in the Venezuelan EEZ")
axis(1, at = seq(1950, 2020, 5)); axis(2, las = 2); box()
legend("topleft", colnames(tab), col = 1:5, pch = 19, lty = 1, lwd = 3, bty = "n", title =

text(2004, 6, cex = 0.8,
     "These Unknown Fishing Country\ncatches appear in the database as an\nindustrial fleet
```



The very substantial increase in international Western Atlantic dolphin catches can be attributed solely to this Unknown Fishing Country catches which appear in the database beginning in 2013. The documentation of the catch reconstruction can be found in the Sea Around Us documentation files (see Mendoza et al. 2015). We spoke to Jeremy Mendoza and other experts who were involved in the reconstruction of domestic catches in Venezuela, and they noted that the "Assigned tuna RFMO catch" actually emerge from spatially reshuffled landing reports to ICCAT, according to the methods of Coulter et al. (2020). In this work, the Sea Around Us team developed a global spatial reallocation of landings of tunas and associated species that were reported to ICCAT. In short, this work uses the fraction of reported spatially disaggregated data to reallocate all data reported to ICCAT. This step is necessary because there are not many countries that report spatial data to ICCAT in the region, and this method may lead to a significant reallocation of landings from the Caribbean to the Venezuelan EEZ, where the local pole and line fishery is significant (J. Mendoza, personal communication). Thus, it is likely that landings from other EEZs, that were included in the reconstructed catch data layer, could have been reassigned to the Venezuelan EEZ.

Because there may be some overlap between the reconstructed data used in the national EEZ estimates, and the ICCAT reporting that was reassigned, there could potentially be a "double counting" of these catch estimates in the Sea Around Us database. These Unknown Fishing Country catches that lead to a large increase in total estimated catches within the Venezuelan EEZ should be treated with caution at this time, until we can verify with Sea Around Us database managers that the spatial disaggregation method used has not led to an overestimate.

3.10 References

Pauly D., Zeller D., Palomares M.L.D. (Editors), 2020. Sea Around Us Concepts, Design and Data (seaaroundus.org).

4 NOAA Fisheries landings data

Here we will explore commercial and recreational landings data accessed through the FOSS (Fisheries One Stop Shop) data portal. The advantage of this portal is that commercial and recreational data are available side-by-side in the same units (pounds landed).

4.1 Data access and upload

Data are downloaded from the NOAA Fisheries FOSS site. For the Data Set, we choose both commercial and recreational. For the Years, we select all years from 1980 to the present. For NMFS Region, we select the "NMFS Regions" button and then select the Gulf, South Atlantic, Middle Atlantic and New England from the box. For the Species, we select "Dolphinfish" and "Dolphinfish**." For Report Format, we choose TOTALS BY YEAR/REGION. Hitting the "RUN REPORT" button then generates the data needed for the subsequent analysis, which can be downloaded as a .csv file under the "Actions" button.

We then read in the data file and clean it for analysis (removing the commas from the numbers so that R recognizes the values as numbers). We output the file header to make sure it was read in and that the numbers were converted correctly.

```
# clear workspace
rm(list = ls())

# read in data file
d <- read.table("data/FOSS_landings.csv", header = T, sep = ",", skip = 1)
head(d)
```

```
  Year     Region.Name   Pounds Dollars   Collection Metric.Tons
1 1980            Gulf   94,984  55,676   Commercial          43
2 1980 Middle Atlantic      400     256   Commercial           0
3 1980     New England      200      54   Commercial           0
```

```
4 1980  South Atlantic    76,042  45,852   Commercial           34
5 1981            Gulf    73,075  46,951   Commercial           33
6 1981            Gulf 1,559,544         Recreational          707
```

```
# take commas out of file --------------------------
d$Pounds <- as.numeric(gsub(",", "", d$Pounds))
d$Dollars <- as.numeric(gsub(",", "", d$Dollars))
d$Metric.Tons <- as.numeric(gsub(",", "", d$Metric.Tons))
head(d)
```

```
  Year     Region.Name  Pounds Dollars   Collection Metric.Tons
1 1980            Gulf   94984   55676   Commercial          43
2 1980 Middle Atlantic     400     256   Commercial           0
3 1980     New England     200      54   Commercial           0
4 1980  South Atlantic   76042   45852   Commercial          34
5 1981            Gulf   73075   46951   Commercial          33
6 1981            Gulf 1559544      NA Recreational         707
```

4.2 Summarize the landings

We then summarize the landings (in pounds) by year, sector, and region. The result is a new data frame with each row representing a year and each column representing the total landings for each region and sector.

```
# compile by year for sectors and regions ---------------
d$Collection <- factor(d$Collection, levels = c("Recreational", "Commercial"))
tab <- tapply(d$Pounds, list(d$Year, d$Collection, d$Region.Name), sum, na.rm = T)
tab[is.na(tab)] <- 0
#head(tab)

dat <- data.frame(cbind(tab[, , "Gulf"], tab[, , "South Atlantic"],
                        tab[, , "Middle Atlantic"], tab[, , "New England"]))
names(dat) <- c("rec_Gulf", "com_Gulf", "rec_SA", "com_SA", "rec_MA", "com_MA", "rec_NE", "
dat
```

```
     rec_Gulf com_Gulf   rec_SA  com_SA  rec_MA com_MA  rec_NE com_NE
1980        0    94984        0   76042       0    400       0    200
1981  1559544    73075  8421898   59037       0      0       0      0
```

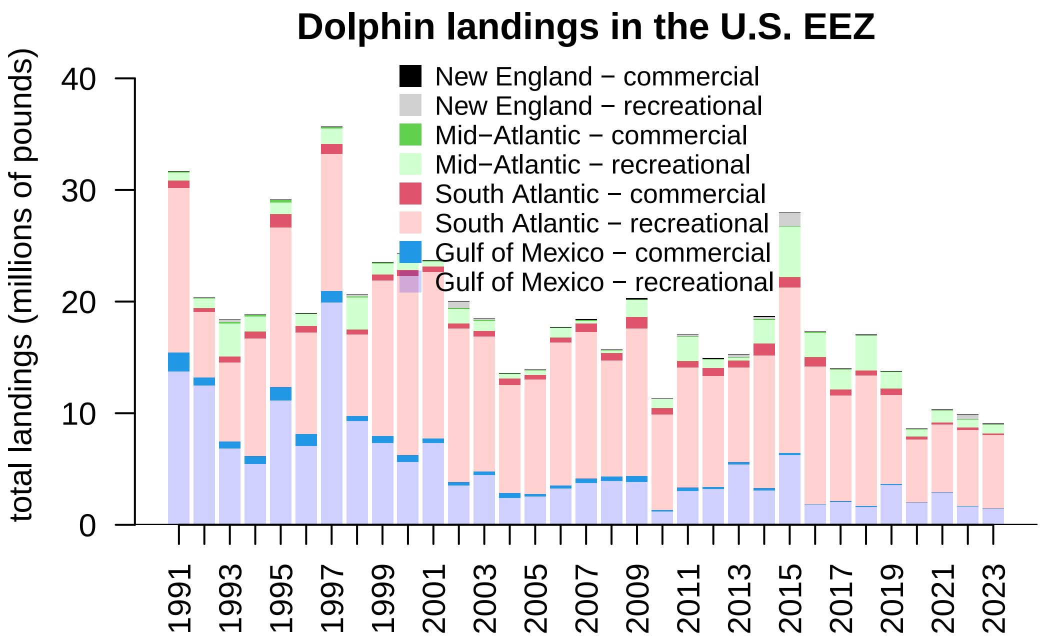
1982	4197503	162457	7573035	143960	2097	200	0	0
1983	2162936	190808	9134022	127986	45927	1400	0	400
1984	1189573	282976	7265430	158901	0	1700	0	400
1985	2542089	269477	6684821	142684	113666	5000	0	4800
1986	5540003	492212	6190187	189747	125653	4200	0	200
1987	4209638	403703	6883546	212905	121433	13400	0	1100
1988	1014312	491610	3134070	212709	130240	26600	2196	17800
1989	12551461	1029628	13273504	426392	1386752	81700	437	15300
1990	13389540	1098819	9283973	514980	624928	69106	2714	14233
1991	13760600	1674124	14756442	650333	722090	90722	6294	9816
1992	12481422	714369	5893700	326831	860112	74423	1411	10315
1993	6873820	596967	7077105	520512	3001335	99420	200927	27069
1994	5448709	679903	10554836	618614	1400635	101044	12586	12328
1995	11142952	1177325	14318412	1164608	1096473	198425	24061	13811
1996	7086006	1047020	9128895	548682	1077364	43581	0	7788
1997	19958088	989648	12284366	877303	1422057	105559	0	13265
1998	9324186	422501	7312090	406368	2920384	86502	120322	11553
1999	7329953	612445	13973364	504263	1005852	98927	4458	5990
2000	5641491	582576	16049049	537077	1433991	36048	0	6545
2001	7355385	368503	14945284	470897	477081	51342	5189	29134
2002	3513226	290601	13794853	436472	1295470	83534	600948	14836
2003	4481937	310836	12112718	459351	929763	46851	129513	21275
2004	2402315	435341	9707678	555973	427145	31916	0	29042
2005	2537132	207695	10299670	376989	437472	19703	0	19982
2006	3290474	224668	12805952	433121	887684	27724	0	19645
2007	3754513	370397	13164377	740255	241212	79832	19542	44707
2008	3937153	384954	10417536	627939	264017	57764	0	16328
2009	3860527	496701	13251102	1007355	1538112	55271	0	92010
2010	1216198	111790	8543542	572031	813738	35869	0	14939
2011	3023914	309324	10774812	543811	2173136	15376	187289	26587
2012	3229090	146256	9983276	704240	755558	44918	26892	46171
2013	5425636	187966	8483569	583556	302421	41936	226988	18564
2014	3074373	201943	11925345	1041856	2129222	58868	183673	61505
2015	6286577	154133	14821772	920454	4501793	36190	1198628	37808
2016	1777348	37411	12357817	828133	2212513	61336	16499	15747
2017	2064106	65141	9457460	532973	1820541	16179	48607	30969
2018	1626095	56188	11713866	435059	3104018	20514	97107	12667
2019	3567170	83956	7967978	563132	1518629	48224	66	16918
2020	1952241	26332	5653294	260311	679556	14040	0	1880
2021	2893817	26570	6061024	178822	1051785	25318	123038	11421
2022	1664771	15906	6845573	182798	708494	17360	466897	9131

```
2023   1427518     13305   6630236   125302   734866   28467   122550    8310
```

4.3 Plot the summary

Now we plot the data, using similar color schemes by region. Because the recreational data have high imprecision prior to the 1990s we plot from 1990 to the last year of data.

```
# select years ---------------------
dat <- dat[which(rownames(dat) > 1990), ]

# define colors and plot -------------------
cols <- c("#0000FF30", 4, "#FF000030", 2, "#00FF0030", 3, "#00000030", 1)

par(mar = c(4, 4, 1, 0), mgp = c(2.5, 1, 0))
yrs <- as.numeric(rownames(dat))

b <- barplot(t(dat)/10^6, names.arg = yrs, col = cols, ylim = c(0, 43),
             border = NA, las = 2, ylab = "total landings (millions of pounds)",
             main = "Dolphin landings in the U.S. EEZ")
legend("top", c("New England - commercial", "New England - recreational",
                "Mid-Atlantic - commercial", "Mid-Atlantic - recreational",
                 "South Atlantic - commercial", "South Atlantic - recreational",
                 "Gulf of Mexico - commercial", "Gulf of Mexico - recreational"),
    col = cols[8:1], pch = 15, pt.cex = 1.5, bty = "n", y.intersp = 0.9, cex = 0.85)
axis(1, at = b, lab = rep("", length(b)), pos = 0)
abline(h=0)
```



4.4 Updated Gulf/Atlantic parsing of data

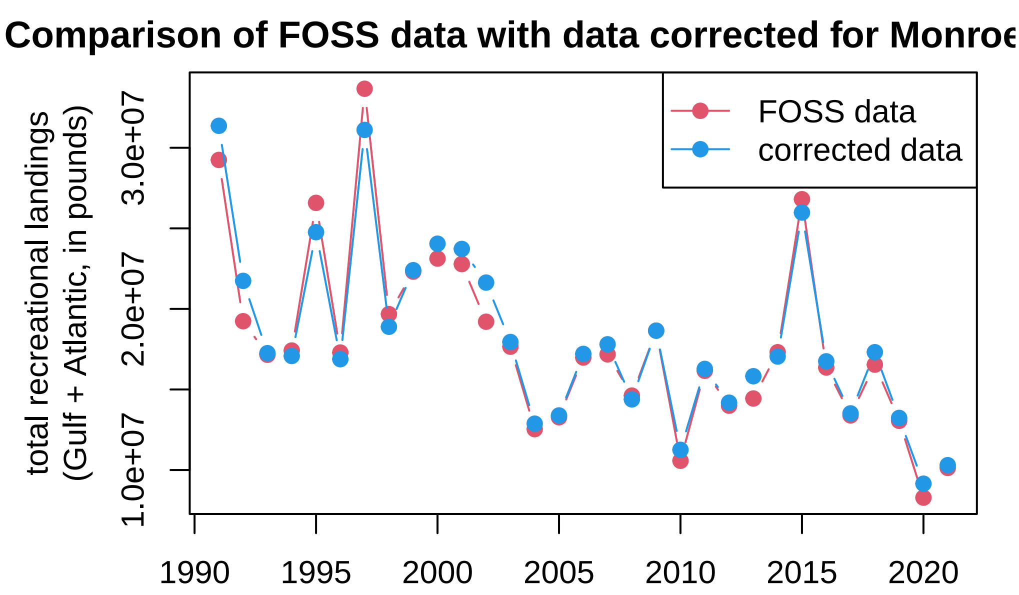
There are some issues with the way that the FOSS system splits out Gulf and Atlantic catches by Monroe County, which is on the border between the two jurisdictions. We received data from Mike Larkin (SERO) which has the correct split out of data for Monroe County between the Gulf and Atlantic. We compare the total recreational catch for the Gulf and Atlantic between the FOSS data and the updated data. The numbers are very close; the average difference from the FOSS data is 2%.

```
ml <- read.table("data/RecreationalDolphinLandings_SAandGOM_MLarkin.csv", header = T, sep =
ml <- ml[which(ml$Year > 1990), ]

# select same years for FOSS data set
dat <- dat[which(yrs <= max(ml$Year)), ]

# check that years match
summary(rownames(dat) == ml$Year)
```

```
   Mode    TRUE
```

```
logical      31
```

```
yrs <- as.numeric(rownames(dat))

# check that totals match
old <- dat$rec_Gulf + dat$rec_SA + dat$rec_MA + dat$rec_NE
new <- ml$GOM + ml$ATL

# plot totals
par(mar = c(4, 6, 2, 1), mgp = c(2.5, 1, 0))
matplot(yrs, cbind(old, new), xlab = "", lty = 1, col = c(2, 4), type = "b", pch = 19,
        ylab = "total recreational landings\n(Gulf + Atlantic, in pounds)",
        main = "Comparison of FOSS data with data corrected for Monroe County")
legend("topright", c("FOSS data", "corrected data"), col = c(2, 4), pch = 19, lty = 1)
```



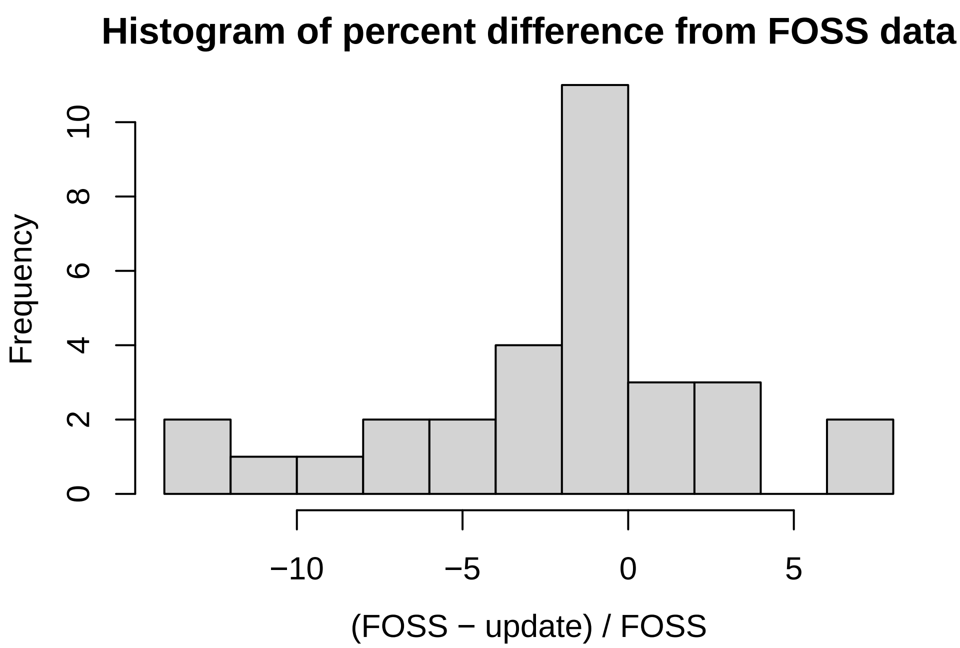

```
# average % difference between FOSS and updated data
mean((old - new)/old*100)
```

```
[1] -2.023886
```

```
# min and max % difference between FOSS and updated data
range((old - new)/old*100)
```

```
[1] -13.008552   7.587789
```

```
hist((old - new)/old*100, main = "Histogram of percent difference from FOSS data", breaks =
     xlab = "(FOSS - update) / FOSS")
```




```
# replace FOSS data with update data
datnew <- dat
datnew$rec_Gulf <- ml$GOM
datnew$rec_SA <- ml$ATL - (datnew$rec_MA + datnew$rec_NE)

# old data set
tail(dat)
```

```
     rec_Gulf com_Gulf   rec_SA com_SA  rec_MA com_MA rec_NE com_NE
2016  1777348    37411 12357817 828133 2212513  61336  16499  15747
2017  2064106    65141  9457460 532973 1820541  16179  48607  30969
```

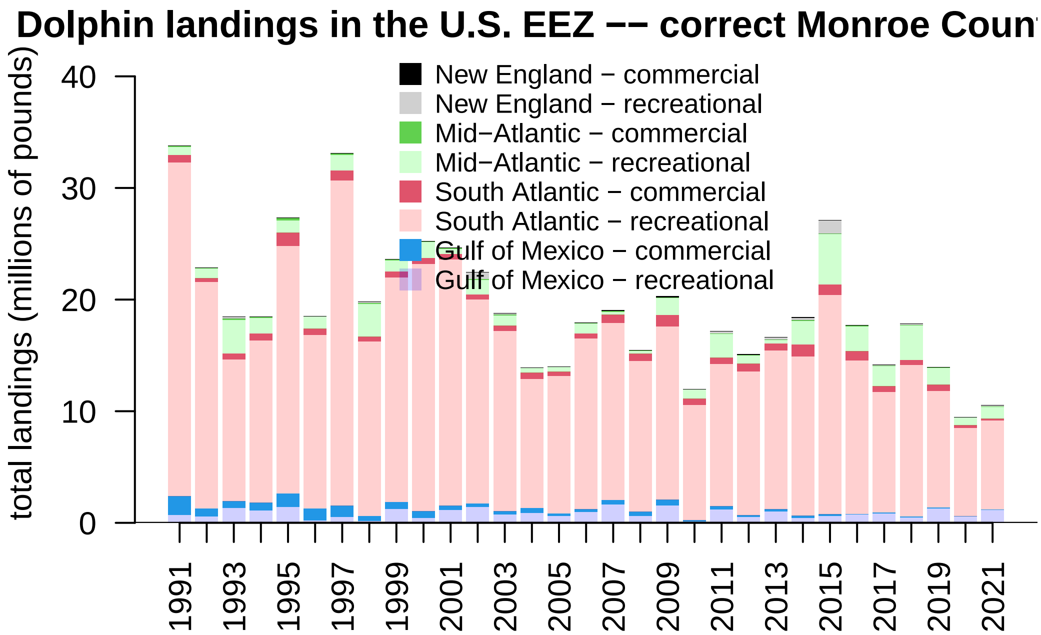
```
2018  1626095    56188 11713866 435059 3104018  20514  97107  12667
2019  3567170    83956  7967978 563132 1518629  48224     66  16918
2020  1952241    26332  5653294 260311  679556  14040      0   1880
2021  2893817    26570  6061024 178822 1051785  25318 123038  11421
```

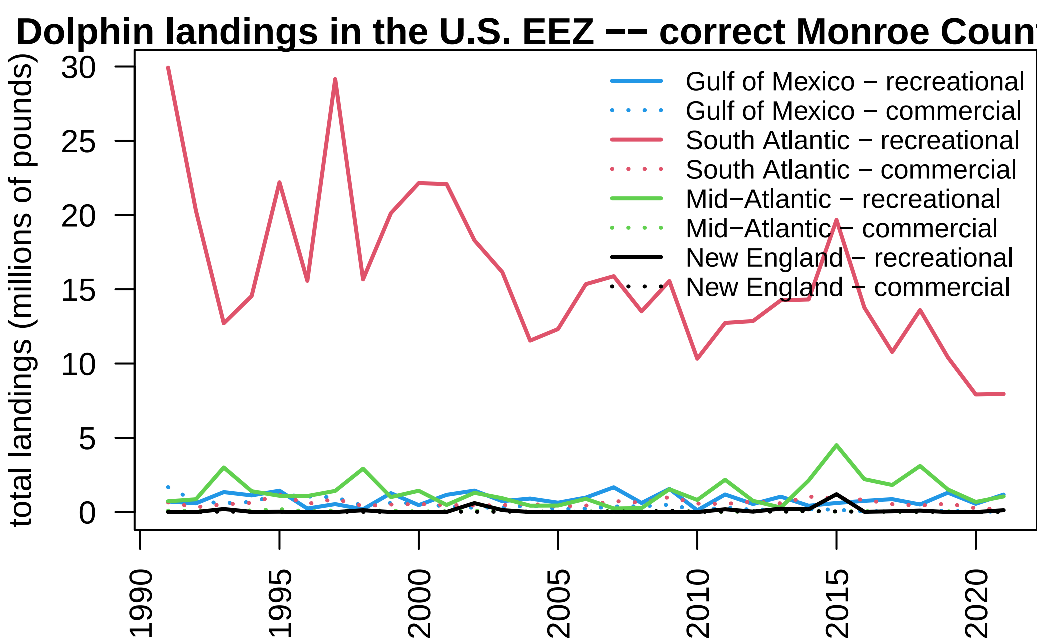
```r
# updated data set
tail(datnew)
```

```
     rec_Gulf com_Gulf   rec_SA com_SA  rec_MA com_MA rec_NE com_NE
2016   751499    37411 13766851 828133 2212513  61336  16499  15747
2017   868489    65141 10777104 532973 1820541  16179  48607  30969
2018   508570    56188 13603875 435059 3104018  20514  97107  12667
2019  1304865    83956 10410599 563132 1518629  48224     66  16918
2020   555695    26332  7916891 260311  679556  14040      0   1880
2021  1174412    26570  7950716 178822 1051785  25318 123038  11421
```

4.5 Plots with updated data

We replace the FOSS values for Gulf and South Atlantic recreational catches with these new values and re-plot the full data set. We can also plot the landings as a line graph so that the relative magnitudes of catch across region and sector are more easily compared.


Dolphin landings in the U.S. EEZ -- correct Monroe Coun
total landings (millions of pounds)
New England − commercial
New England − recreational
Mid−Atlantic − commercial
Mid−Atlantic − recreational
South Atlantic − commercial
South Atlantic − recreational
Gulf of Mexico − commercial
Gulf of Mexico − recreational


Dolphin landings in the U.S. EEZ -- correct Monroe Coun
total landings (millions of pounds)
Gulf of Mexico − recreational
Gulf of Mexico − commercial
South Atlantic − recreational
South Atlantic − commercial
Mid−Atlantic − recreational
Mid−Atlantic − commercial
New England − recreational
New England − commercial

5 SST

6 Analysis of ocean temperature trends

Coming soon